Double-Interaction (Pileup) Effects in PPG12: Efficiency, Shower Shapes, Truth-Vertex Reweighting, Cluster Size, and Energy Response

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Abstract

Run-24 $p+p$ data at $\sqrt{s} = 200\,\mathrm{GeV}$ contain two crossing-angle periods (0 mrad and 1.5 mrad) with very different pileup conditions, where two independent collisions in the same bunch crossing shift the reconstructed vertex and distort cluster kinematics, shower shapes, and ABCD templates. This note consolidates the PPG12 treatment of double interactions (DI) across four lines of evidence: (i) luminosity-weighted MBD-Poisson pileup fractions (22.4% cluster-weighted at 0 mrad, 7.9% at 1.5 mrad); (ii) the DI efficiency study, which decomposes the apparent 94% → 31% double-MC drop into an 88% matching artifact (removed by switching from ΔR-based to track-ID-only matching) and 12% genuine cluster loss, leaving a 9.4% relative efficiency reduction (52.0%/57.4%) on the mixed-0 mrad integrated photon-ID chain and a 2.8% integrated cross-section impact at 1.5 mrad (reco 1.6% + BDT 1.0% + iso 0.2%); (iii) an iterative data-driven truth-vertex reweight $w(z)$ factorized as $w(z_h)\,w(z_{\mathrm{mb}})$, which closes the individual DI truth-vertex distributions that the production $f(z_r)$ structurally cannot; (iv) data-vs-mixed-MC shower-shape validation with up to 20.8× χ^2/ndf improvements at 0 mrad for vertex-sensitive variables. The nominal recipe is: single-MC efficiency with track-ID matching, the iterative truth-vertex reweight on both periods, mixed MC for shower-shape validation only. The +1.4% 1.5 mrad cross-section bias under $f(z_r)$ alone is removed inside the $|z_h| < 60\,\mathrm{cm}$ fiducial, with per-bin closure at 1–2%.

Contents

1 Introduction

In $p+p$ collisions at RHIC, two separate collisions can occur in the same bunch crossing (pileup, also referred to here as double interaction, DI). For PPG12's isolated prompt-photon cross-section measurement at $\sqrt{s} = 200$ GeV using sPHENIX Run 24 data, pileup is not just a rate correction: it shifts the reconstructed MBD vertex to the average of the two collision points, which in turn distorts cluster pseudorapidity via the EMCal-radius geometric mapping, biases the truth-reco matching that underlies the efficiency calculation, broadens w_η^{CoG} at the shower-shape level, enters the ABCD background templates through DI-broadened jet clusters, and — if uncorrected — biases the truth fiducial acceptance used for the efficiency denominator at the percent level.

The analysis uses two crossing-angle periods (split at run 51274):

- **0 mrad**, runs 47289–51274, $\mathcal{L} \simeq 32.7\,\text{pb}^{-1}$, broad luminous region ($\sigma_z \simeq 54$ cm), high pileup ($\sim$22% cluster-weighted DI);
- **1.5 mrad** (nominal published period), runs 51274–54000, $\mathcal{L} \simeq 16.9\,\text{pb}^{-1}$, Piwinski-narrowed luminous region ($\sigma_z \simeq 22$ cm), modest pileup ($\sim$8% cluster-weighted DI).

The PPG12 DI programme consists of two complementary work streams:

1. **Full-GEANT DI MC blending**: dedicated `photon10_double` and `jet12_double` samples, produced via full detector simulation of two overlapping $p+p$ collisions, are mixed with the nominal single-interaction samples at the physics fractions of Table 1. This blend provides both the DI efficiency study (§5) and the shower-shape data–MC validation (§6).
2. **Truth-vertex reweighting**: the production reco-level weight $f(z_r) = D(z_r)/M(z_r)$ is structurally insufficient to close the individual truth vertices of double-interaction events; an iterative data-driven reweight $w(z)$ applied per truth vertex ($w(z_h)$ for single MC, $w(z_h)\,w(z_{\text{mb}})$ for doubles) is introduced in §7 and removes the +1.4% cross-section bias at 1.5 mrad.

A third, toy-level vertex-shift simulation (`DoubleInteractionCheck.C`) exists for cross-checks on single MC and is described in `.claude/rules/double-interaction.md`; this note focuses exclusively on the full-GEANT and truth-vertex-reweight work streams that feed the nominal analysis.

The remainder of the note covers: pileup fractions (§2), DI MC production (§3), the vertex-shift mechanism (§4), the efficiency impact with the full DI decomposition (§5), shower-shape data–MC validation (§6), truth-vertex reweighting (§7), cluster size under DI (§8), energy response (§9), eta-edge migration (§10), and the summary (§11). Appendices cover the DI reco–truth kernel, discarded reweighting methods, the baseline TruthVertexRecoCheck study, and a file/code inventory.

2 Pileup Fractions

The mean number of interactions per bunch crossing, μ, is derived from the Poisson-corrected MBD coincidence rate:

$$\mu = \frac{\sum_{\text{runs}} N_{\text{MBD,corr}}}{r_B \cdot \sum_{\text{runs}} \Delta t}, \tag{1}$$

where $r_B = 111 \times 78{,}200 = 8.68 \times 10^6$ Hz is the RHIC bunch-crossing rate. The event-level interaction fractions under the zero-truncated Poisson are

$$f_1 = \frac{\mu\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_2 = \frac{(\mu^2/2)\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_{\geq 3} = 1 - f_1 - f_2. \tag{2}$$

Fractions are computed per run and luminosity-weighted (not simple-averaged) to avoid Jensen's inequality bias. Cluster-weighted fractions account for the higher cluster multiplicity of multi-interaction events:

$$w_{\text{single}} = \frac{f_1}{\sum_k k\, f_k}, \qquad w_{\text{double}} = \frac{2\, f_2 + 3\, f_{\geq 3}}{\sum_k k\, f_k}, \tag{3}$$

with triple-and-higher interactions folded into the double bin for the purposes of the GEANT blend (the `photon10_double` / `jet12_double` productions model the two-collision case exactly; extra collisions are rare enough that this approximation is well within the bin-by-bin closure tolerance).

Table 1: Luminosity-weighted pileup fractions and cluster-level mixing weights from `calc_pileup_range.C`. Triple+ folded into f_2 for blending. μ_{corr} is the MBD-corrected mean interactions per bunch crossing.

Crossing angle	Run range	μ_{corr}	f_1	f_2	w_{single}	w_{double}
0 mrad	47289–51274	0.24	0.879	0.111	0.776	**0.224**
1.5 mrad	51274–54000	0.08	0.960	0.039	0.921	**0.079**

The cluster-weighted fractions w_{double} (last column of Table 1) are the numbers used throughout this note: 0.224 at 0 mrad and 0.079 at 1.5 mrad. They multiply the per-event cross-section weight of each sample in the full-GEANT blending pipeline (§6.1) and set the double-interaction mixing fraction f_d in the iterative truth-vertex forward model (§7.3).

3 DI MC Production

The full-GEANT DI samples used throughout this note are `photon10_double` (photon signal, truth $p_T^{\gamma} \in [10, 100]$ GeV with window $[14, 30]$ GeV dominant) and `jet12_double` (QCD jet, $p_T^{\text{jet}} \in [14, 21]$ GeV), produced in the run28 production area `anatreemaker/macro_maketree/sim/run28/`. The double-interaction events are generated via the Fun4All embedding flag 2 (pythia-in-hijing), carrying a primary hard-scatter vertex (z_h) and a secondary minimum-bias vertex (z_{mb}). Both vertex values are available in the output slimtree through the branches `vertexz_truth` (z_h) and `vertexz_truth_second` (z_{mb}, read in C++ as `vertexz_truth_mb`).

Known production caveats. Two independent pipeline reviews (`condor_pipeline_review_run28.md`, findings #7 and #19; the BDT training audit of Apr 2026) flagged the following issues that the reader should be aware of:

- **Hardcoded `total_line=10000` on the pre-submit stage**: the per-sample `run_condor.sh` templates in 12 of 14 sample directories hardcode a 10,000-line filelist cut-off, while `photon10_double` (9998 lines) and `jet12_double` (9997) sit just below the cutoff. The final 2–3 jobs per double sample therefore read empty filelists and produce empty `caloana.root` files that are silently merged into the combined output by `hadd_combined.sh`. Two jet samples (`jet5`, `jet8`) already patched this with a dynamic `wc -l` look-up; the remaining samples have not.

- **Missing `vertexz_truth_mb` branch**: the `photon10_double` and `jet12_double` `bdt_split.root` files currently downstream of `apply_BDT.C` were produced before the `vertexz_truth_mb` branch was added to the anatreemaker output. Consumers downstream (notably `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C`, which both carry an optional-pointer guard) silently drop the secondary-vertex factor $w(z_{\text{mb}})$ for those two samples. The impact on cluster-size means and shower-shape moments is expected to be below the percent level at 1.5 mrad (where the DI blend fraction is only 7.9%) and sub-percent at 0 mrad (where the MC beam profile is already close to the luminous region). Regenerating these two `bdt_split.root` files from the reprocessed `combined.root` (which does carry `vertexz_truth_mb`) removes this caveat; it is a follow-up.

- **Embedding flag buried in 3 of 14 Fun4All macro copies**: `TruthJetInput::add_embedding_flag(2)` is set in the Fun4All macros of `photon10_double`, `jet12_double`, and `jet8` (for distinct physics reasons); any consolidation of the 14-way macro duplication should preserve this flag explicitly on the DI productions.

None of these issues blocks the DI programme at its current precision; the cross-section bias from the missing $w(z_{\rm mb})$ factor is sub-percent and well below the existing systematic budget at 1.5 mrad. The pipeline consolidation is tracked separately in `condor_pipeline_review_run28.md`.

4 Vertex-Shift Mechanism

In a double-interaction event the MBD reconstructs a single vertex whose best estimator is the average of the two collision points:

$$z_{\rm vtx}^{\rm reco} \approx \frac{z_{\rm vtx,1} + z_{\rm vtx,2}}{2}. \tag{4}$$

The RHIC luminous region has $\sigma_{\rm lum} \approx 22$ cm at 1.5 mrad and $\sigma_{\rm lum} \approx 54$ cm at 0 mrad, while the MC generator was produced with a wider $\sigma_{\rm gen} \approx 65$ cm common to both periods. The two vertices in a DI event are drawn from the generator distribution (MC) or from the luminous region (data), so the displacement between the primary truth vertex and the reconstructed vertex, $\Delta z \equiv z_{\rm vtx}^{\rm reco} - z_{\rm vtx}^{\rm truth}$, can reach several tens of cm in either case. The cluster pseudorapidity shift induced at the EMCal-barrel radius $R_{\rm EMCal} = 93.5$ cm follows directly from the cluster geometry:

$$\Delta\eta \approx -\frac{\Delta z}{R_{\rm EMCal}\cosh\eta}. \tag{5}$$

At $\eta = 0$ this reduces to $|\Delta\eta| \approx |\Delta z|/93.5$ cm, so a matching cut of $\Delta R < 0.1$ fails whenever $|\Delta z| \gtrsim 9.35$ cm at midrapidity. This is the critical observation that drives the efficiency interpretation in §5: pileup events populate $|\Delta z|$ far beyond this threshold.

Table 2: Vertex-shift distributions in single- and double-interaction photon10 MC. The critical threshold $|\Delta z| = 9.35$ cm maps onto $|\Delta\eta| = 0.1$ at $\eta = 0$ via Eq. (5).

Quantity	Single MC	Double MC
RMS of Δz [cm]	7.6	34.5
Fraction with $\|\Delta z\| > 9.35$ cm	—	83%

The matching-failure rate inherits the same deterministic geometric origin, saturating to 100% for vertex displacements well past the 9.35 cm threshold:

Table 3: $\Delta R < 0.1$ matching-failure rate vs. $|\Delta z|$ in double-interaction MC.

$\|\Delta z\|$ range [cm]	Matching-failure rate
< 10	9%
10–15	77%
15–20	∼100%
> 20	100%

Figure 1 shows the same quantity as a continuous curve in single-interaction MC, where the rare tail of large $|\Delta z|$ events gives a clean step at the geometric threshold.

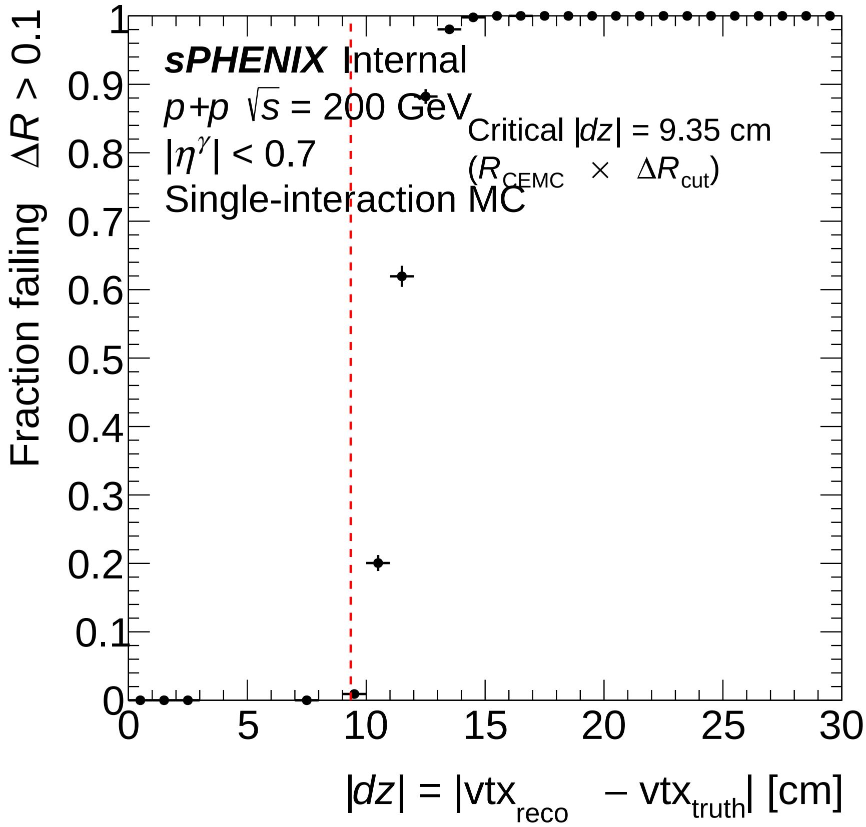


Figure 1: $\Delta R < 0.1$ failure fraction as a function of $|\Delta z|$ in single-interaction MC. The sharp step at $\sim$9.35 cm matches the prediction from Eq. (5), confirming vertex mismatch as the sole driver of ΔR matching failure; cluster quality is essentially unchanged across the step.

5 Efficiency Impact

5.1 Apparent Efficiency Drop Under ΔR Matching

When the `photon10_double` sample was first passed through the legacy efficiency pipeline (which required a $\Delta R < 0.1$ geometric match on top of the GEANT track-ID to associate a reconstructed cluster with a truth photon), the reconstruction efficiency dropped dramatically: from $\sim 94\%$ in single-interaction MC to $\sim 31\%$ in the double-interaction sample. Taken at face value this would imply that pileup destroys two-thirds of the reconstructed cluster sample — a result in obvious tension with the $\sim 1\%$ fiducial-acceptance bias expected from the pileup fraction alone.

A dedicated investigation showed that the apparent drop was overwhelmingly a *truth–reco matching artifact* caused by the vertex-displacement mechanism of §4. The single decisive observation is that the matching failure rate scales as expected from Eq. (5): in double MC, 83% of events have $|\Delta z| > 9.35$ cm, of which 77%+ fail the $\Delta R < 0.1$ cut. The cluster is physically present at the correct track-ID; the reconstructed η is merely offset from the true η by the vertex shift. Switching the matching strategy from ΔR-geometric to track-ID-only recovers those clusters and eliminates the artifact. This section decomposes the apparent drop, justifies the removal of $\Delta R < 0.1$ from the production pipeline, and reports the factorized per-stage efficiency under the corrected matching.

5.2 Decomposition: Matching Artifact vs. Genuine Cluster Loss

For truth photons that failed the $\Delta R < 0.1$ match in the double MC, the failure mode was classified by searching for a cluster with the correct `trkID` irrespective of ΔR:

- **ΔR failure (88%):** a cluster with the correct `trkID` exists, but sits at $\Delta R > 0.1$ due to vertex displacement. The signal-photon cluster is fully reconstructed and carries the right energy; it simply appears at the wrong reconstructed η when the cluster position is taken with respect to the displaced MBD vertex.
- **Missing cluster (12%):** no cluster carries the matching `trkID`. This is genuine cluster loss — the photon shower merged into a neighbouring jet shower or was absorbed by the higher-multiplicity environment.
- **Extra truth photons ($\sim 0\%$):** the second collision does not produce additional truth-level prompt photons at the level relevant for PPG12. A per-event check on 100k photon10 events shows 0.493 truth photons/event in single MC and 0.495 in double MC (statistically identical); the ≥ 2-denominator fraction is 0.22% vs. 0.21%; see Table 4.

Table 4: Truth-level prompt photon content in photon10 samples (100k events each). Denominator: $|\eta^\gamma| < 0.7$, $E_\mathrm{T}^{\mathrm{iso,truth}} < 4\,\mathrm{GeV}$. The slight increase in mean truth $E_\mathrm{T}^{\mathrm{iso,truth}}$ in double MC (0.26 vs. 0.17 GeV) reflects additional hadrons from the second collision falling within the isolation cone, but remains well below the 4 GeV fiducial threshold.

Quantity	Single MC	Double MC
Mean truth photons per event	0.493	0.495
Events with ≥ 2 denom. photons	0.22%	0.21%
Mean truth $E_\mathrm{T}^{\mathrm{iso,truth}}$ [GeV]	0.17	0.26

The dominant 88% matching-artifact component is entirely driven by vertex-shifted kinematics (a geometric reconstruction effect, not a physics effect), and is removed by switching to track-ID matching. The residual 12% is genuine cluster-loss and is kept as a genuine pileup degradation in the reconstruction-stage efficiency.

5.3 Baseline Rejection on Single MC (ΔR Cut is Never Benign)

A secondary question is whether $\Delta R < 0.1$ matters outside of pileup at all. It does: even in single-interaction MC, $\Delta R < 0.1$ rejects $\sim 3.5\%$ of correctly `trkID`-matched truth photons (Table 5), and the rejection is driven entirely by vertex mismatch ($|\Delta z| > 9.35\,\mathrm{cm}$) rather than cluster quality: 73% of rejected clusters have $E_\mathrm{T}/p_\mathrm{T}^{\mathrm{truth}} \in [0.8, 1.2]$, i.e. a nearly-correct reconstructed energy. Six physics arguments support the removal — the track-ID is ground truth; the ΔR cut tests vertex resolution rather than calorimeter performance; the response matrix already absorbs vertex-induced kinematic shifts; and the $\sim 3\%$ bias is z-dependent and therefore propagates into any period-dependent quantity. Track-ID-only matching is therefore adopted as the nominal both for the efficiency and for the DI-MC study.

Table 5: $\Delta R < 0.1$ failure rates in single-interaction MC. "Fail ΔR" is the fraction of `trkID`-matched photons rejected by the geometric cut.

Sample	Truth photons	No trkID match	Fail $\Delta R < 0.1$ (of matched)
photon5	273,017	3.42%	2.70%
photon10	2,380,500	2.37%	2.98%
photon20	2,840,703	1.72%	4.01%
Combined	5,494,220	2.09%	3.50%

5.4 Per-Stage Efficiency Under Track-ID Matching

With $\Delta R < 0.1$ removed, the factorized efficiency chain $\varepsilon_{\mathrm{reco}} \to \varepsilon_{\mathrm{iso|reco}} \to \varepsilon_{\mathrm{id|reco+iso}} \to \varepsilon_{\mathrm{all}}$ reveals the genuine pileup effects stage by stage. Table 6 presents the integrated efficiency for single, double, and mixed (0 mrad) photon10 MC under `trkID`-only matching; Figure 2 shows the same information as a function of truth p_{T}.

Table 6: Per-component efficiency with `trkID`-only matching (no ΔR cut), photon10 samples integrated over $E_{\mathrm{T}}^{\gamma} = 14$–$30$ GeV. Mixed $= 0.776 \times$ single $+ 0.224 \times$ double (0 mrad cluster-weighted fractions from Table 1).

Efficiency component	Single MC	Double MC	Mixed 0 mrad	Effect (single $\to$ double)
$\varepsilon_{\mathrm{reco}}$ (reconstruction)	93.5%	73.1%	88.4%	-20 pp (cluster loss + shifted kinematics
$\varepsilon_{\mathrm{iso}}$ (isolation | reco)	74.2%	71.0%	73.5%	-3 pp (extra cone energy)
$\varepsilon_{\mathrm{id}}$ (BDT | reco+iso)	82.7%	69.4%	80.0%	**−13 pp** (vertex-shifted kinematics)
$\varepsilon_{\mathrm{all}}$ (total)	57.4%	36.0%	52.0%	-21 pp (reco + BDT dominate)

Three structural observations:

- **Reconstruction** ($\varepsilon_{\mathrm{reco}}$): the 20 pp gap between single (93.5%) and double (73.1%) MC splits into genuine cluster loss (no-`trkID`-match rate: 7.9% double vs. 2.4% single, i.e. ∼5.5 pp of the drop) and common-cut failures from shifted kinematics (∼15 pp).
- **Isolation** ($\varepsilon_{\mathrm{iso}}$): essentially unchanged (3 pp degradation). Extra cone energy from the second collision is present but small compared with the 4 GeV isolation threshold.
- **Photon ID** ($\varepsilon_{\mathrm{id}}$): a **13 pp degradation** in double MC (69.4% vs. 82.7%), entirely driven by vertex-displaced shower-shape inputs to the parametric E_{T}-dependent BDT threshold. In single MC the same effect is negligible (-0.2 pp) because only ∼2.3% of clusters have large vertex shifts. In double MC, ∼58% of recovered clusters are "new" (had $\Delta R > 0.1$), and these have a BDT pass rate of only 59.8% vs. 82.7% for the full population.

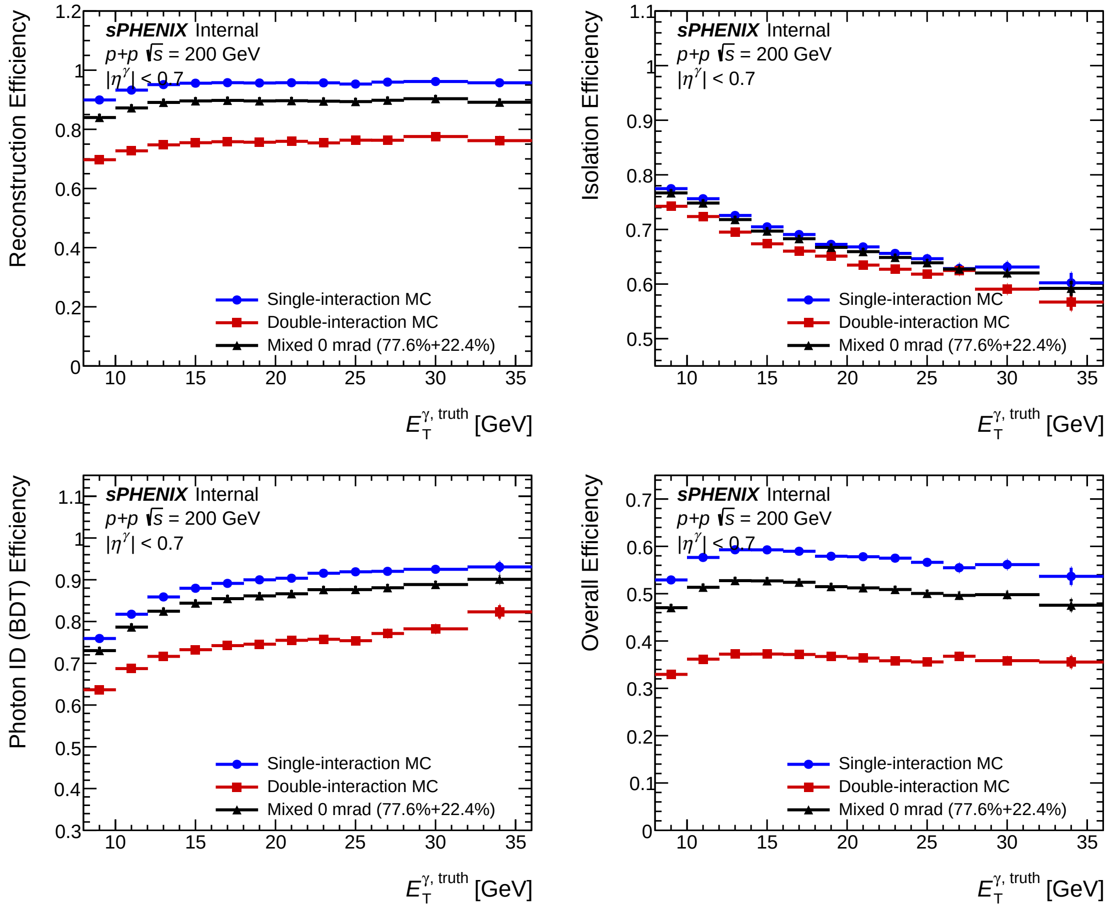


Figure 2: Per-stage efficiency vs. truth p_{T} without the ΔR cut, single (blue), double (red), and mixed 0 mrad (black) photon10 MC. Upper left: reconstruction. Upper right: isolation. Lower left: photon ID (BDT). Lower right: total. The BDT stage shows the cleanest 13 pp gap between single and double MC, driven by vertex-shifted kinematics feeding an E_{T}-dependent BDT threshold.

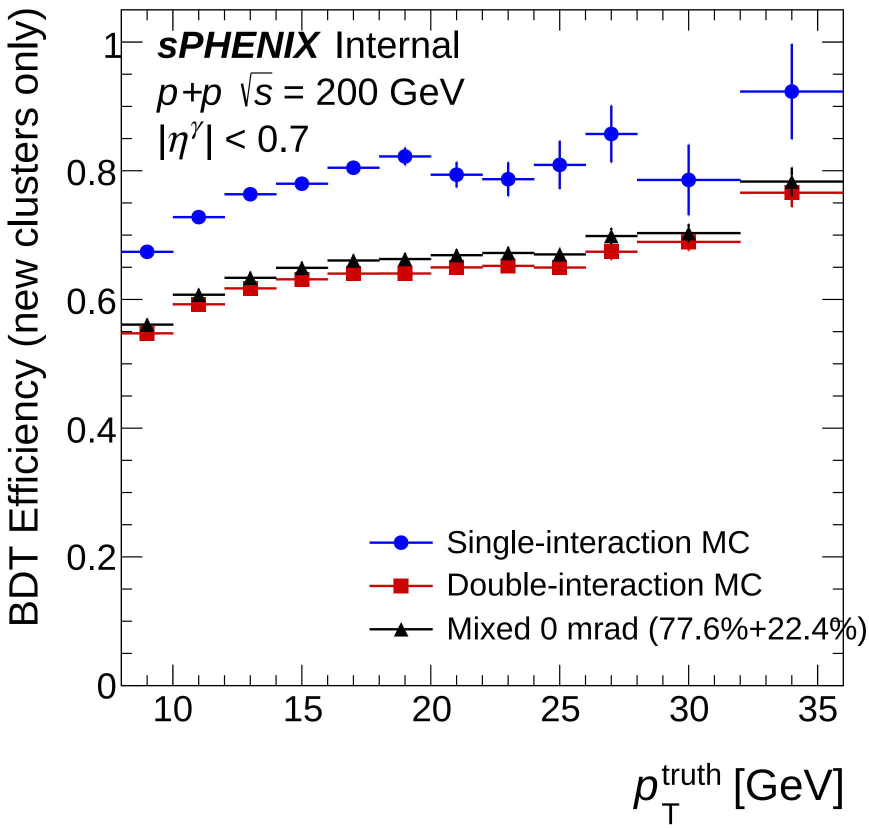


Figure 3: BDT pass rate for "new" clusters (those with $\Delta R > 0.1$ but valid `trkID`) vs. truth p_{T}. These vertex-shifted clusters have 10–13 pp lower BDT pass rates than the full population, with the deficit narrowing at higher p_{T} where the parametric BDT score is further above threshold.

For reference, Table 7 shows the legacy efficiency with $\Delta R < 0.1$ retained. The large reco drop there is the matching artifact; the BDT appears "unaffected" only because vertex-shifted clusters have already been pre-filtered out of the denominator.

Table 7: Per-component efficiency under the legacy $\Delta R < 0.1$ match, for reference. The reco drop is the matching artifact; the BDT only looks unaffected because vertex-shifted clusters have been silently rejected upstream.

Efficiency component	Single MC	Double MC	Mixed 0 mrad
$\varepsilon_{\mathrm{reco}}$ ($\Delta R < 0.1$)	91.4%	30.7%	76.2%
$\varepsilon_{\mathrm{iso}}$	74.2%	71.1%	73.9%
$\varepsilon_{\mathrm{id}}$ (BDT)	82.9%	82.6%	82.9%
$\varepsilon_{\mathrm{all}}$	56.2%	18.0%	46.6%

5.5 `TEfficiency` Weighted-Event Bug Fix

During the ΔR investigation a bug was identified in the `TEfficiency` usage in `RecoEffCalculator_TTreeReader` the previous `SetWeight(w) + Fill(pass, pT)` pattern stored unweighted event counts, and `hadd`-ing across samples would therefore combine them by raw event count rather than by cross-section weight. The corrected implementation is `SetUseWeightedEvents()` plus `FillWeighted(pass, weight, pT)`, which stores weighted sums and composes correctly under `hadd`. All numbers in Table 6 use the corrected implementation.

5.6 Implications for the Nominal Analysis

- **Reco loss contributes** 1.6% **to the 1.5 mrad cross section**: 20 pp single→double efficiency drop times the 7.9% cluster-weighted pileup fraction.
- **BDT degradation contributes** 1.0%: 0.079×13 pp.
- **Isolation contributes** 0.2%: 0.079×3 pp.
- **Total integrated DI effect on the 1.5 mrad cross section**: $1.6\% + 1.0\% + 0.2\% =$ **2.8**% pre-correction, partially correlated with the vertex-shift systematic that is already evaluated separately. At the total-efficiency level (mixed 0 mrad), the same decomposition gives $\varepsilon_{\text{all}}^{\text{mix}}/\varepsilon_{\text{all}}^{\text{single}} = 52.0/57.4$, i.e. a $1 - 52.0/57.4 =$ **9.4**% relative reduction (Table 6). The 1.5 mrad effect is smaller than this because the pileup fraction is $\sim 3\times$ smaller than at 0 mrad.

The nominal analysis uses **single-interaction MC for the efficiency calculation** (with `trkID`-only matching). Mixed MC is reserved for the shower-shape data–MC validation (§6) and the ABCD background-template cross-check. Double MC is not used as a default because the apparent bulk efficiency drop is dominated by the matching artifact rather than a genuine physics effect: closing it would require simulating the low-pileup target precisely rather than averaging a broad generator profile.

6 Shower-Shape Data-vs-Mixed-MC Validation

6.1 Two-Pass Blending Pipeline

The pileup blend is built with `run_showershape_double.sh` in two passes over `ShowerShapeCheck.C`. The first pass computes a blended MC vertex-z template, the second runs the full shower-shape analysis using it:

Pass 1 (vertex scan). `ShowerShapeCheck.C` is called with `do_vertex_scan=true` and fills only the z_{vtx} histogram, weighting each sample by its cluster-level fraction (w_{single} or w_{double} from Table 1). Five parallel jobs run: `photon10_double(0.224)`, `jet12_double(0.224)`, `photon10_nom(0.776)`, `jet12_nom(0.776)`, and data. The output vtxscan ROOT files are merged per-period via `hadd` to form the blended MC vertex distribution.

Pass 2 (full analysis). `ShowerShapeCheck.C` is re-run with `mix_weight` (multiplicative fraction applied to all histogram fills) and `vtxscan_sim_override` (path to the blended vtxscan file from Pass 1) set. All MC samples use the *same* blended vtxscan for vertex reweighting, which ensures consistent z_{vtx} templates across samples regardless of which sample originally provided the vertex distribution. Each sample's fills are scaled by its mixing weight on top of the cross-section weight. Single and double outputs are `hadd`-merged to produce the final blended MC.

The full signature is

```
ShowerShapeCheck(configname, filetype, doinclusive, do_vertex_scan, mix_weight, vtxscan_sim_override)
```

with `mix_weight` = 0.224 (or 0.776) at 0 mrad and 0.079 (or 0.921) at 1.5 mrad, and `vtxscan_sim_override` pointing at the Pass-1 blended ROOT file. The same script accepts a single command-line argument overriding the default 0 mrad fraction, so running the 1.5 mrad blend is simply `bash run_showershape_double.sh config_showershape.yaml 0.079`.

Truth-vertex reweighting. Starting from the April 15 regeneration, the shower-shape blending pipeline applies the iterative truth-vertex reweight (§7) via the `truth_vertex_reweight_on` / `truth_vertex_reweight_file` fields in the showershape configs, rather than the reco-level $f(z_r)$ alone. The samples shown in this section use this nominal setting.

6.2 Comparison Methodology

Shower-shape distributions are compared at two selection levels:

- **cut0**: no selection. Data includes NPB contamination that is not modeled in MC, which complicates interpretation.
- **cut1**: after the NPB preselection (NPB score > 0.5). This removes non-physics clusters and provides the cleanest pileup assessment. **All plots in this section use cut1.**

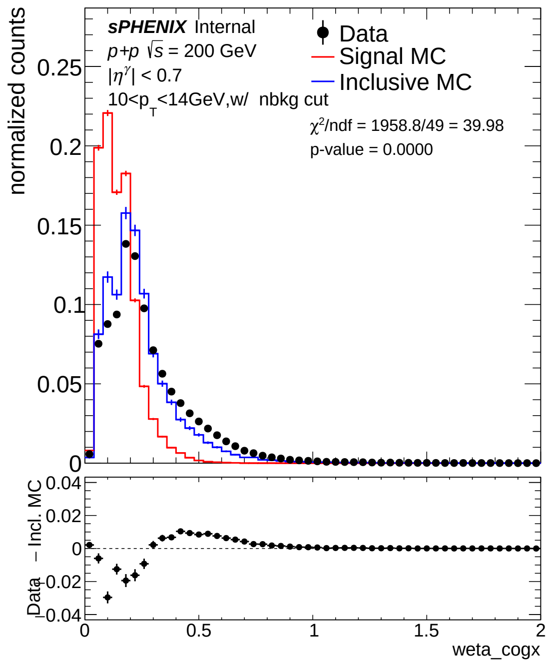
Each comparison plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with a χ^2/ndf and p-value annotation and a Data − Inclusive MC difference panel. Left panels show nominal (single-only) MC; right panels show mixed (double-blended) MC.

6.3 0 mrad Period (22.4% Double Interaction)

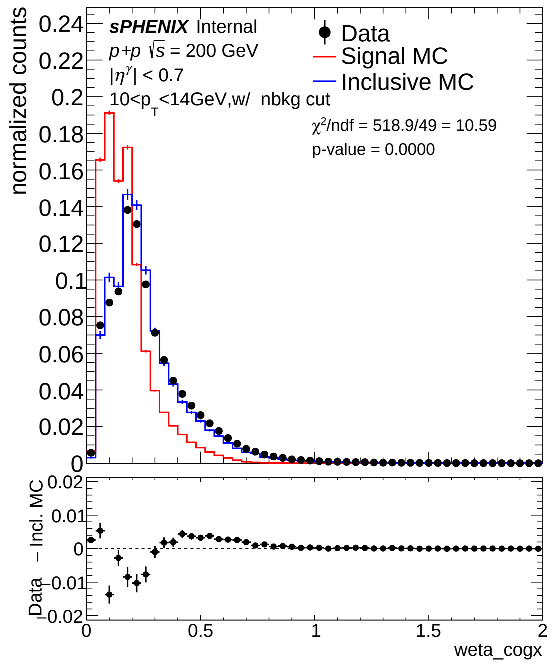
At 0 mrad the pileup fraction is high and the nominal (single-only) MC misrepresents the data at the cut0/cut1 shower-shape level for pileup-sensitive variables. Mixing in 22.4% of DI MC closes the gap substantially for w_η^{CoG} at $E_\mathrm{T} \geq 14\,\mathrm{GeV}$.

6.3.1 w_η^{CoG} (eta cluster width)

w_η^{CoG} is the variable most sensitive to DI: vertex shifts along the beam axis broaden the reconstructed η profile directly via Eq. (5).

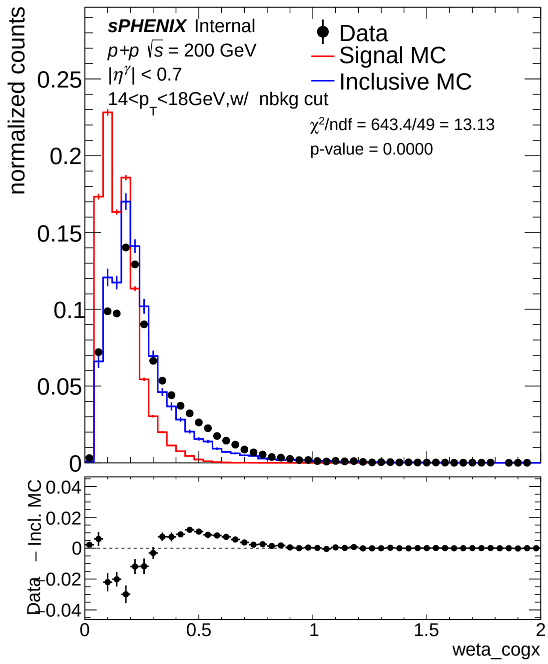


(a) Nominal MC

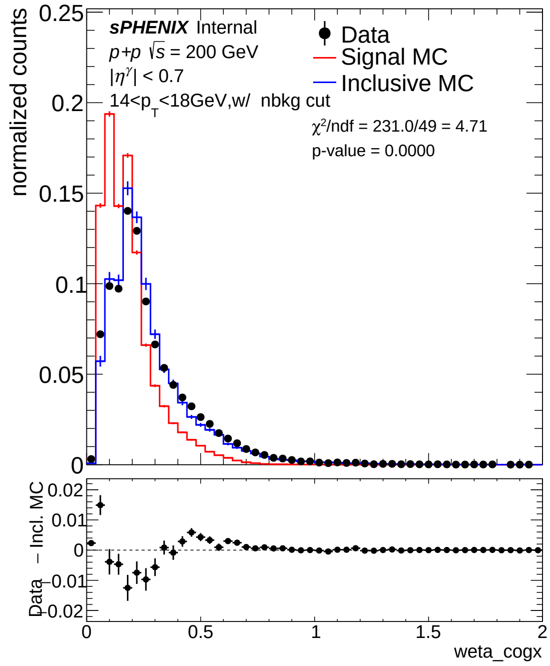


(b) Mixed MC (22.4% DI)

Figure 4: w_η^{CoG}, $10 < E_\mathrm{T} < 14\,\mathrm{GeV}$ (pt0), 0 mrad, cut1. Nominal χ^2/ndf = 47.40; mixed χ^2/ndf = 28.96 (1.6× improvement).

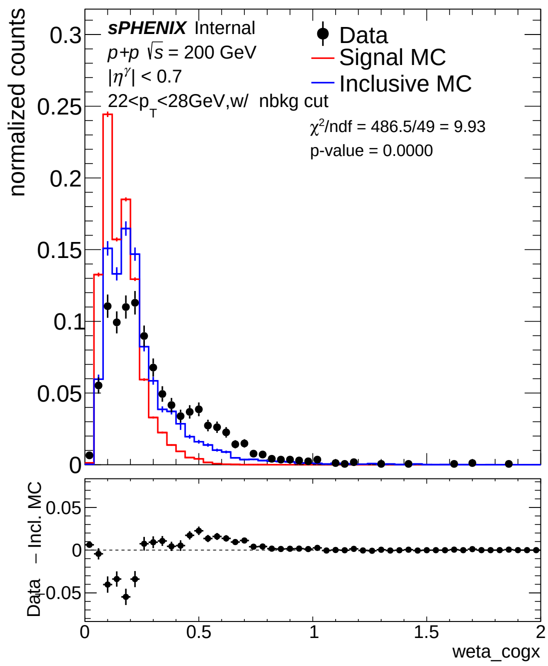


(a) Nominal MC

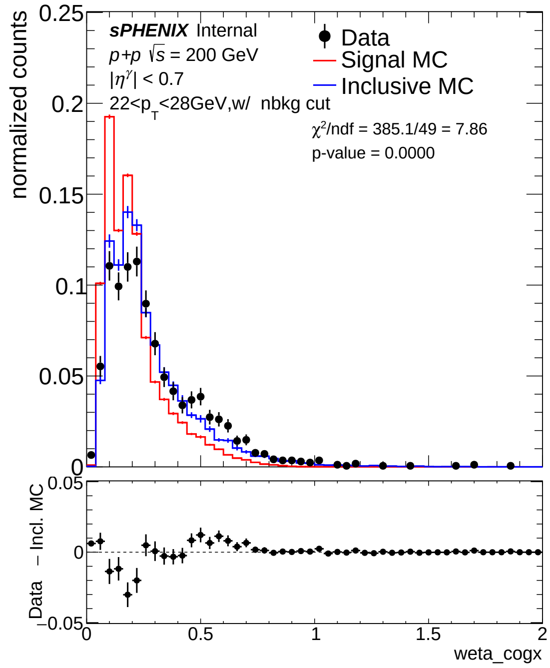


(b) Mixed MC (22.4% DI)

Figure 5: w_η^{CoG}, $14 < E_{\mathrm{T}} < 18\,\mathrm{GeV}$ (pt1), 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 17.64$; mixed $\chi^2/\mathrm{ndf} = 2.99$ ($5.9\times$ improvement).



(a) Nominal MC



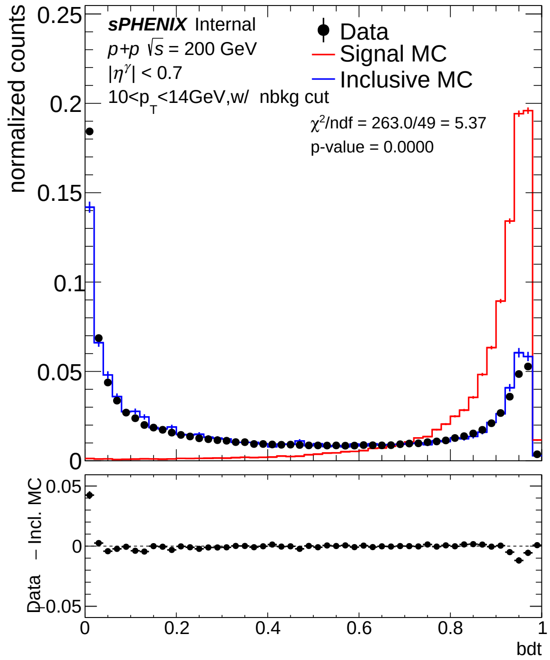
(b) Mixed MC (22.4% DI)

Figure 6: w_η^{CoG}, $22 < E_{\mathrm{T}} < 28\,\mathrm{GeV}$ (pt3), 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 15.58$; mixed $\chi^2/\mathrm{ndf} = 0.75$ ($p = 0.84$; $20.8\times$ improvement). The mixed MC achieves $p = 0.84$, i.e. the data-MC residual in this bin is consistent with statistical fluctuation.

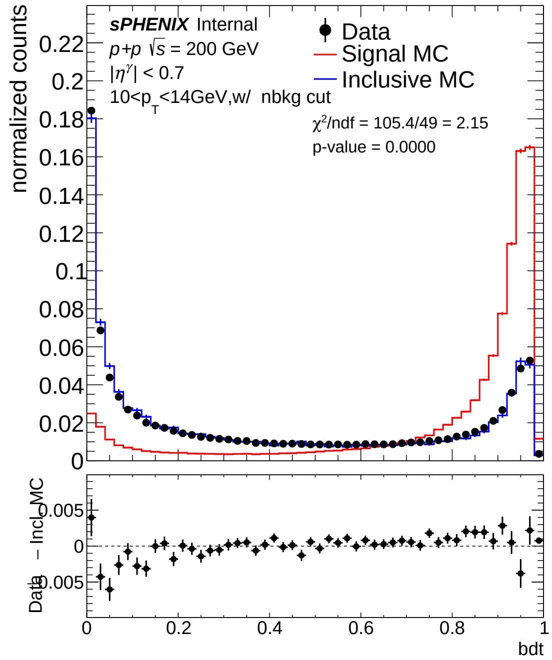
The improvement grows with E_{T}: at 22–28 GeV the nominal MC discrepancy becomes large relative to statistical uncertainties, and the mixed MC provides the largest relative correction.

6.3.2 BDT score

The photon/jet BDT amplifies underlying shower-shape distortions through its nonlinear decision function and is the most sensitive composite variable at the cluster level.

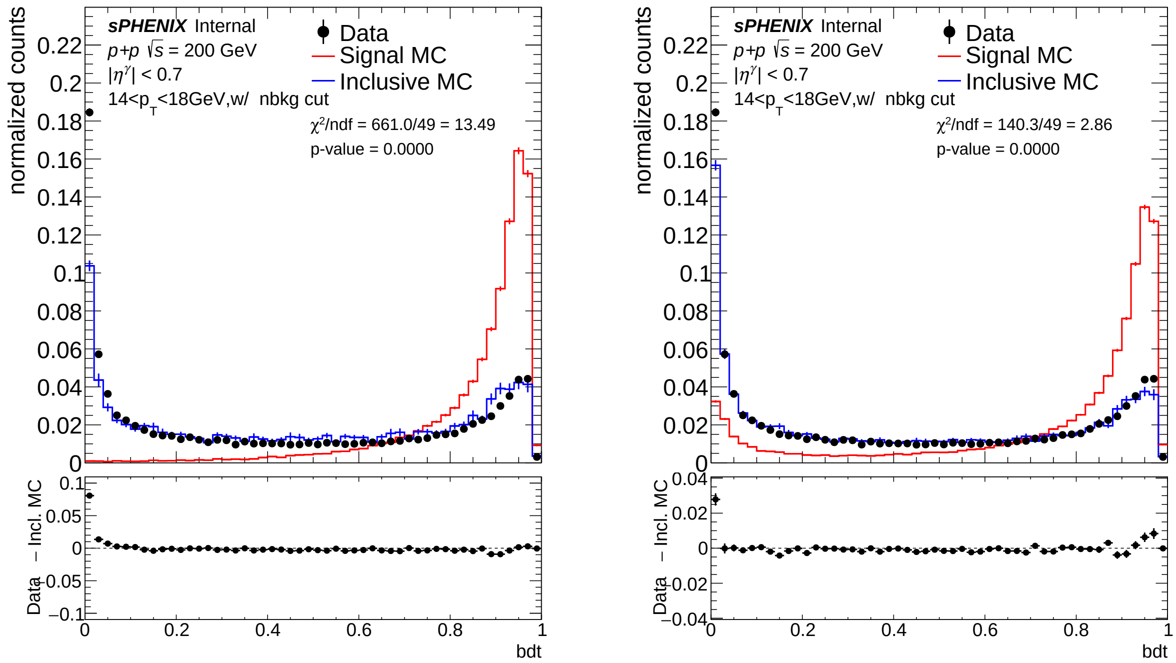


(a) Nominal MC



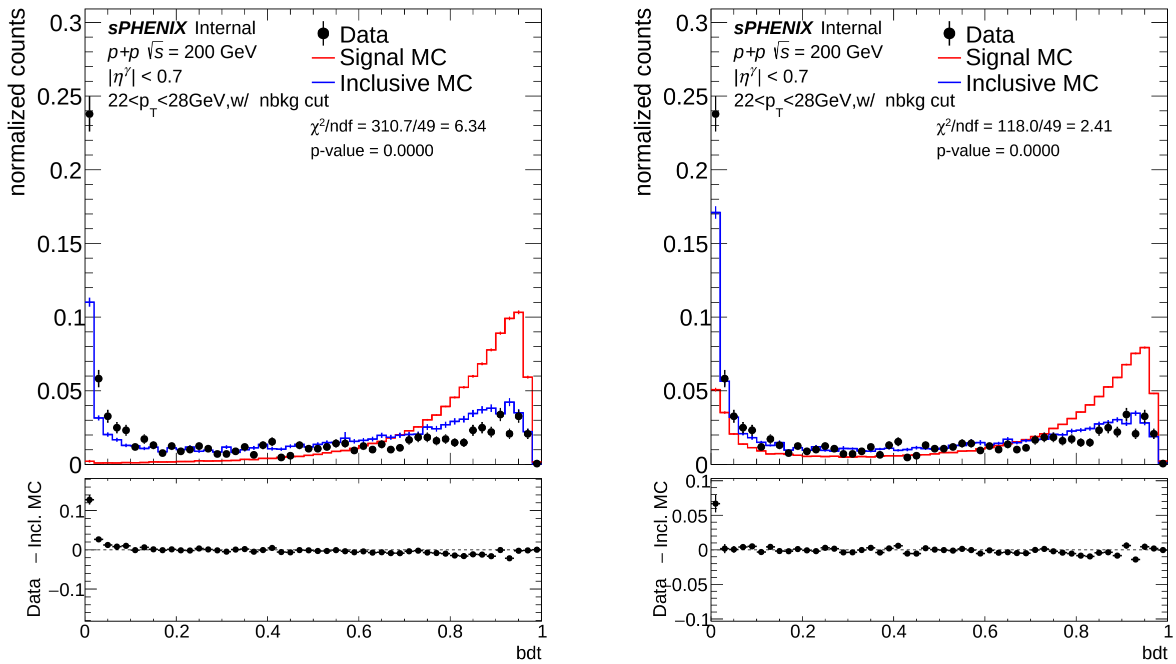
(b) Mixed MC (22.4% DI)

Figure 7: BDT score, $10 < E_T < 14$ GeV (pt0), 0 mrad, cut1. Nominal $\chi^2/\text{ndf} = 6.85$; mixed $\chi^2/\text{ndf} = 20.30$ (3.0× **degradation**). See §6.7 for the discussion of the low-E_T crossover.



(a) Nominal MC (b) Mixed MC (22.4% DI)

Figure 8: BDT score, $14 < E_{\mathrm{T}} < 18\,\mathrm{GeV}$ (pt1), 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 12.20$; mixed $\chi^2/\mathrm{ndf} = 1.63$ ($p = 0.004$; $7.5\times$ improvement).

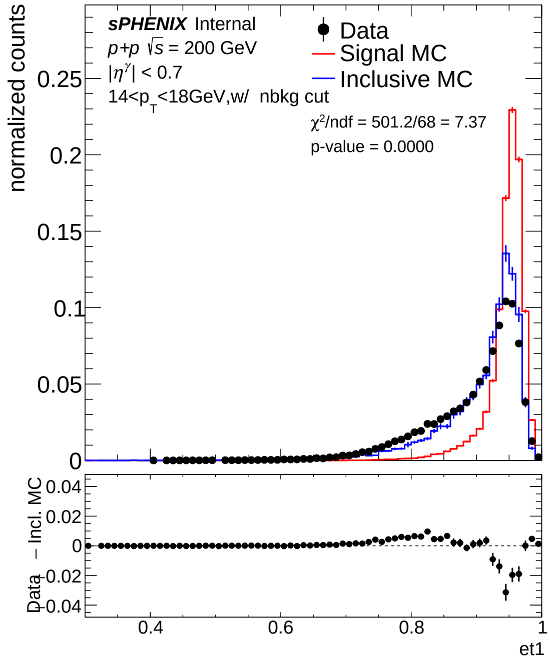


(a) Nominal MC (b) Mixed MC (22.4% DI)

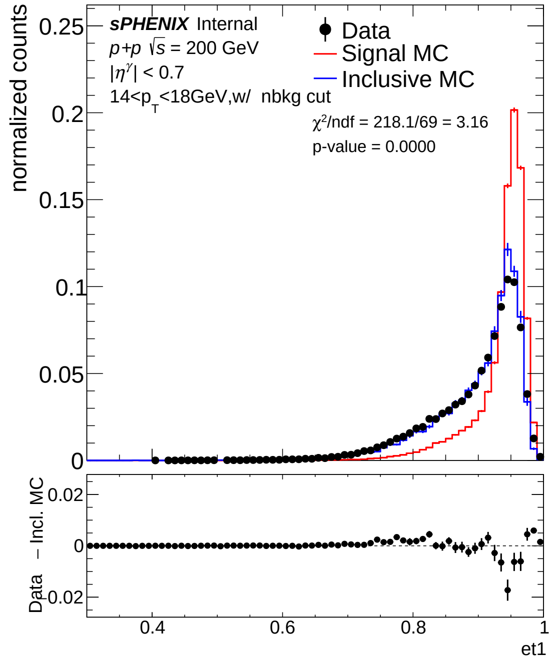
Figure 9: BDT score, $22 < E_{\mathrm{T}} < 28\,\mathrm{GeV}$ (pt3), 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 9.62$; mixed $\chi^2/\mathrm{ndf} = 1.75$ ($p = 0.001$; $5.5\times$ improvement).

6.3.3 f_1 and $E_{1\times1}/E_{3\times3}$

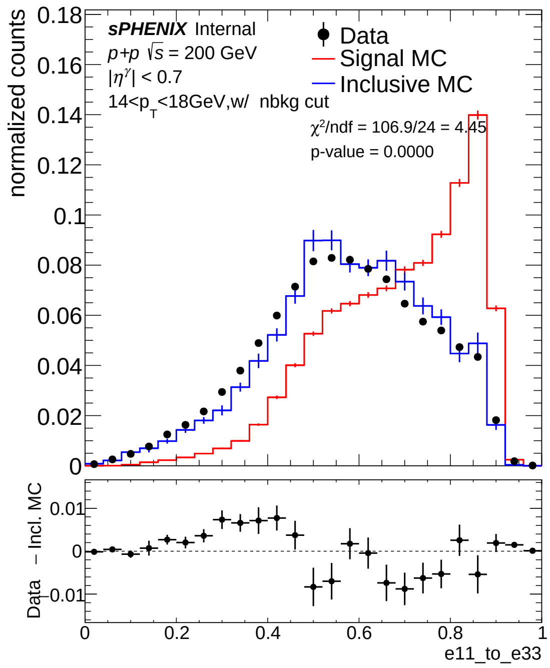
Double interactions spread energy outward from the shower core, reducing the inner-ring energy fraction f_1. The $E_{1\times1}/E_{3\times3}$ ratio is less sensitive to pileup broadening.



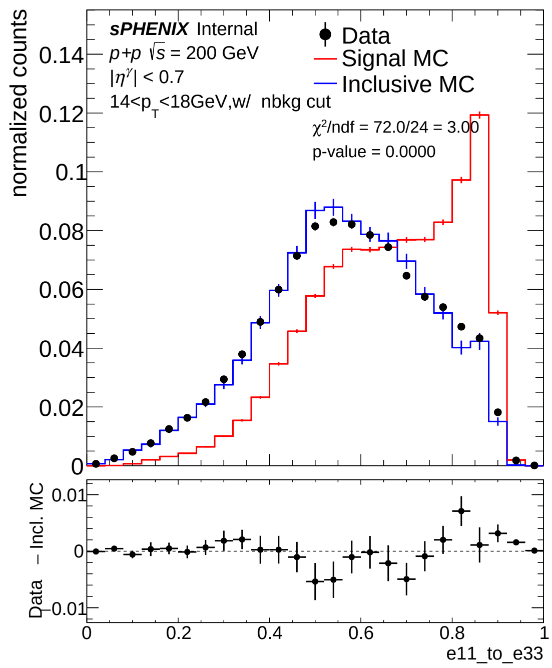
(a) f_1, nominal MC



(b) f_1, mixed MC (22.4% DI)



(c) $E_{1\times1}/E_{3\times3}$, nominal MC



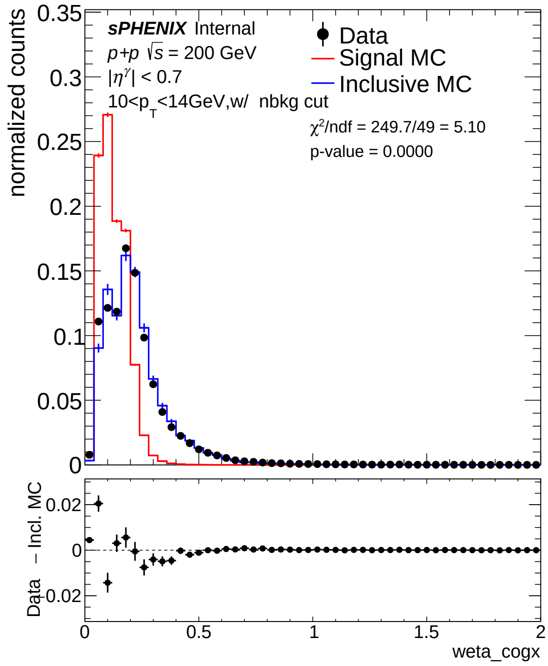
(d) $E_{1\times1}/E_{3\times3}$, mixed MC (22.4% DI)

Figure 10: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18$ GeV (pt1), 0 mrad, cut1. f_1: $\chi^2/\mathrm{ndf} = 9.33 \to 1.92$ (4.9×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\mathrm{ndf} = 4.50 \to 4.14$ (1.1×).

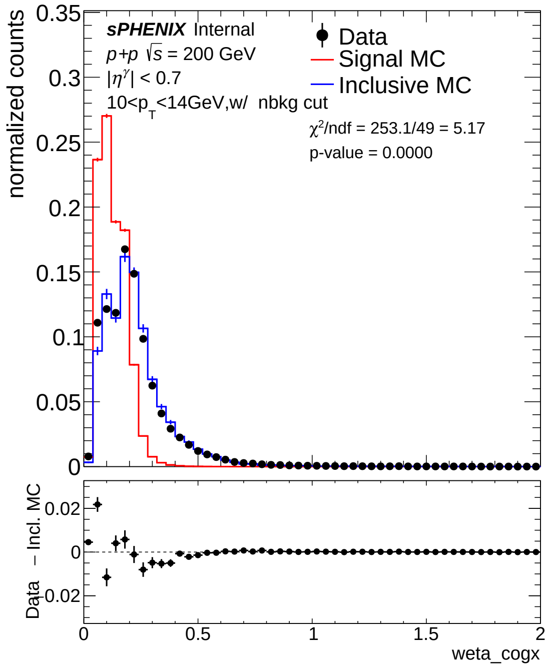
6.4 1.5 mrad Period (7.9% Double Interaction)

At 1.5 mrad the DI fraction is $\sim 3\times$ smaller than at 0 mrad, so the mixed MC correction is a few times smaller. The nominal agreement at cut1 is already good for most variables (χ^2/ndf $\lesssim 10$); where mixed MC helps, the improvement is mainly at the high-E_{T} bins.

6.4.1 w_η^{CoG}

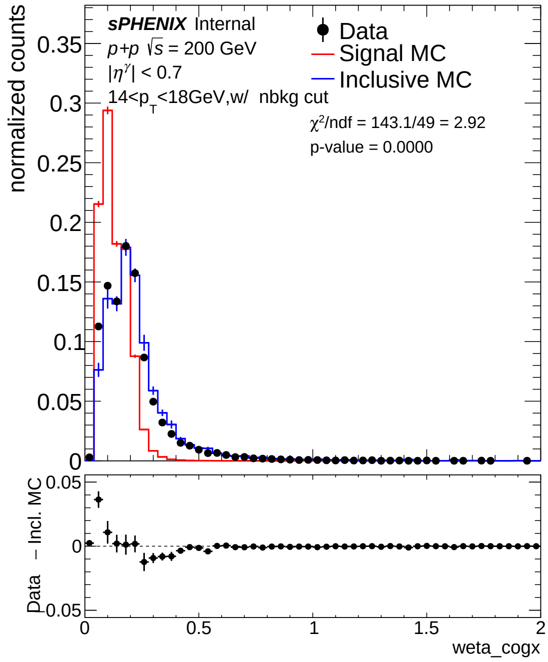


(a) Nominal MC

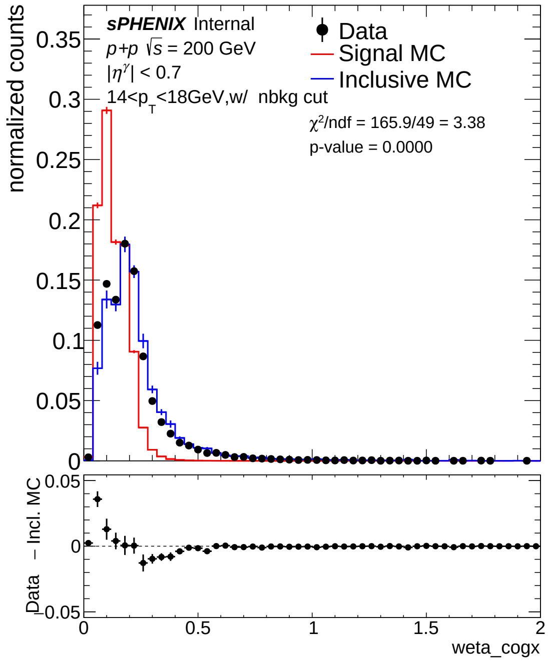


(b) Mixed MC (7.9% DI)

Figure 11: w_η^{CoG}, $10 < E_{\mathrm{T}} < 14$ GeV (pt0), 1.5 mrad, cut1. Nominal χ^2/ndf $= 9.72$; mixed χ^2/ndf $= 42.31$ (4.4$\times$ **degradation**).

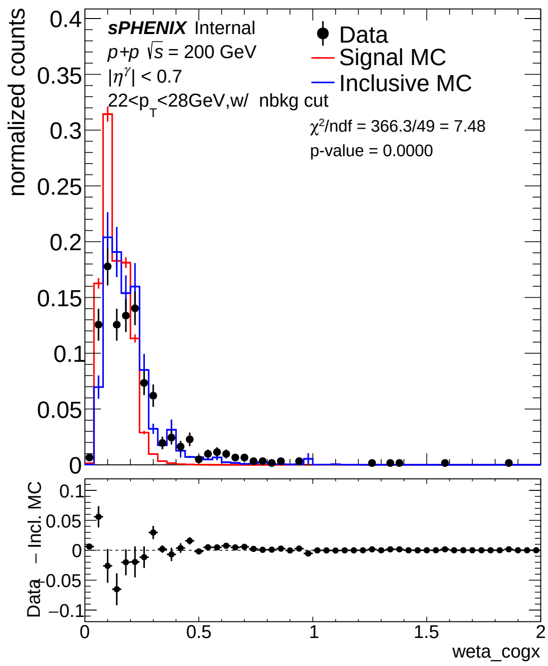


(a) Nominal MC

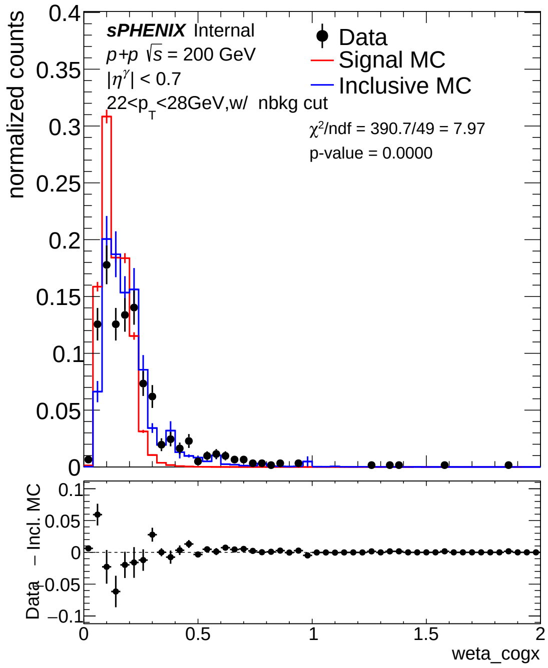


(b) Mixed MC (7.9% DI)

Figure 12: w_η^{CoG}, $14 < E_{\mathrm{T}} < 18\,\mathrm{GeV}$ (pt1), 1.5 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 6.73$; mixed $\chi^2/\mathrm{ndf} = 4.86$ (1.4× improvement).



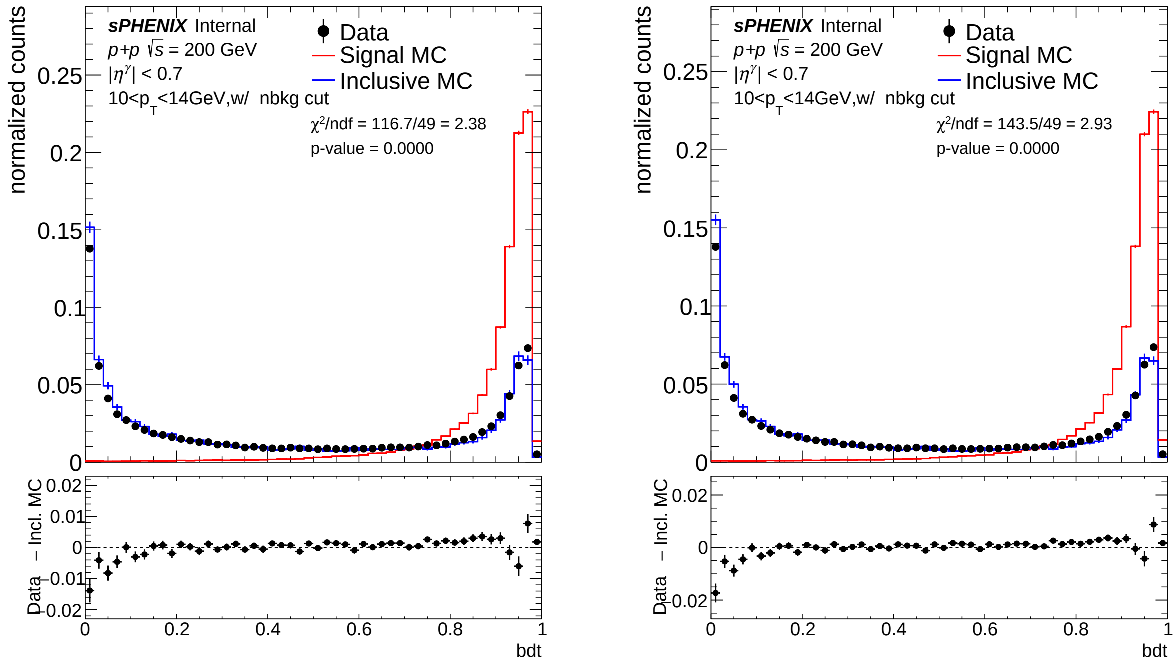
(a) Nominal MC



(b) Mixed MC (7.9% DI)

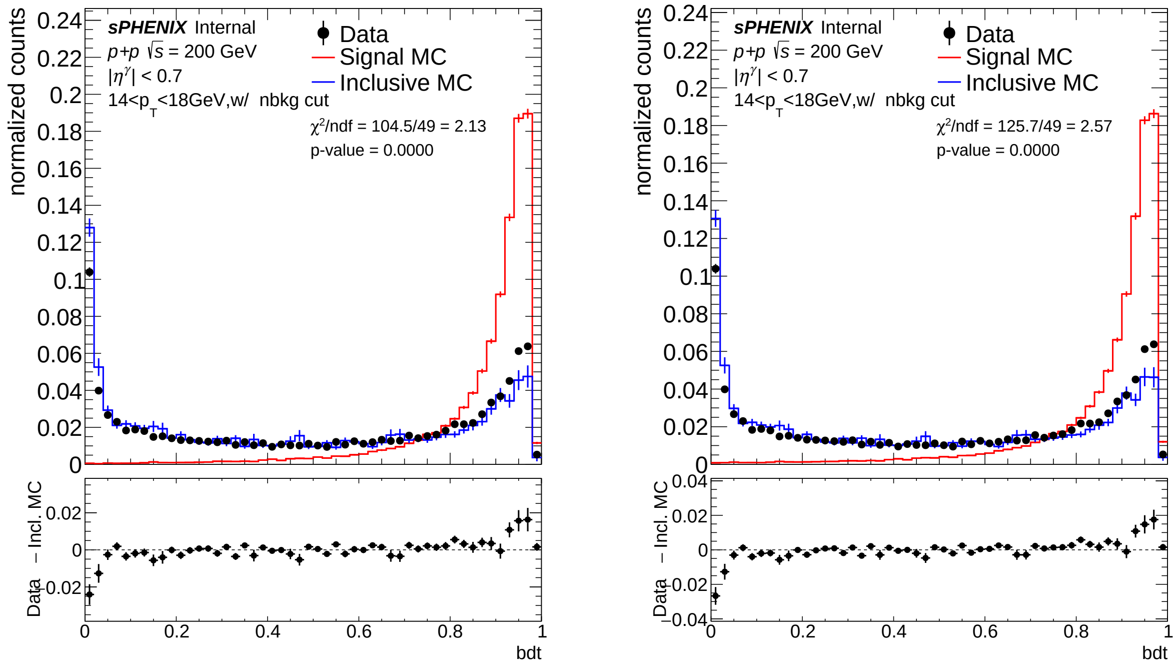
Figure 13: w_η^{CoG}, $22 < E_{\mathrm{T}} < 28\,\mathrm{GeV}$ (pt3), 1.5 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 21.10$; mixed $\chi^2/\mathrm{ndf} = 1.31$ ($p = 0.118$; 16.1× improvement).

6.4.2 BDT score



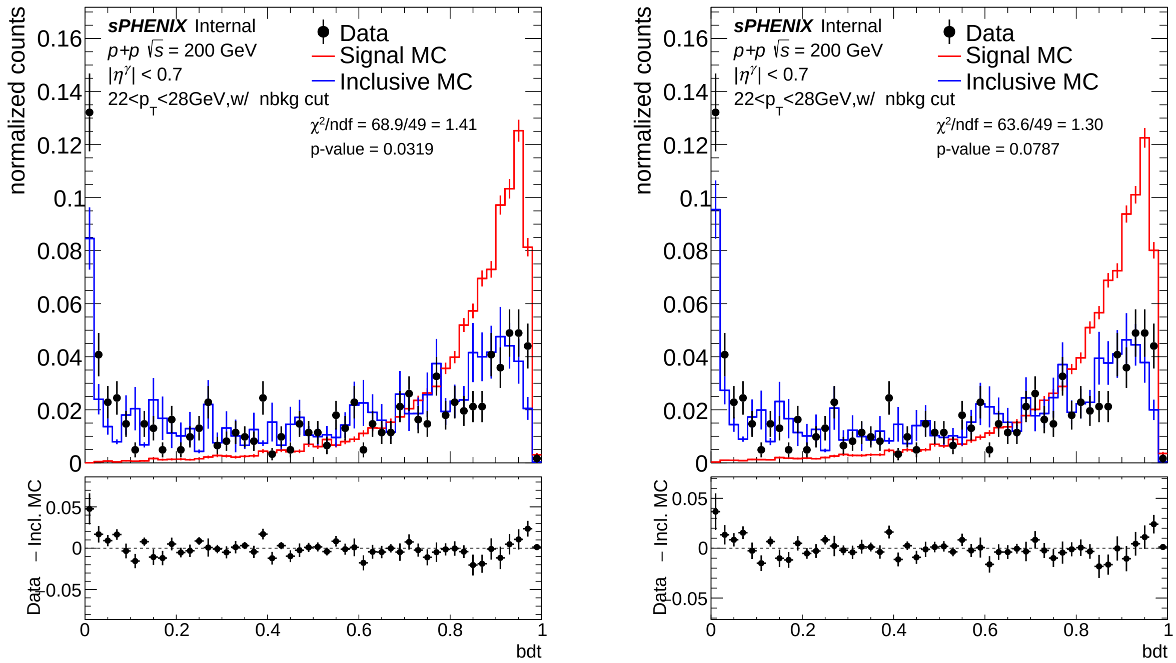
(a) Nominal MC (b) Mixed MC (7.9% DI)

Figure 14: BDT score, $10 < E_{\mathrm{T}} < 14\,\mathrm{GeV}$ (pt0), 1.5 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 5.98$; mixed $\chi^2/\mathrm{ndf} = 33.08$ (5.5× **degradation**).



(a) Nominal MC (b) Mixed MC (7.9% DI)

Figure 15: BDT score, $14 < E_{\mathrm{T}} < 18\,\mathrm{GeV}$ (pt1), 1.5 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 1.16$ ($p = 0.207$); mixed $\chi^2/\mathrm{ndf} = 2.31$ (2.0× **degradation**; nominal already excellent).

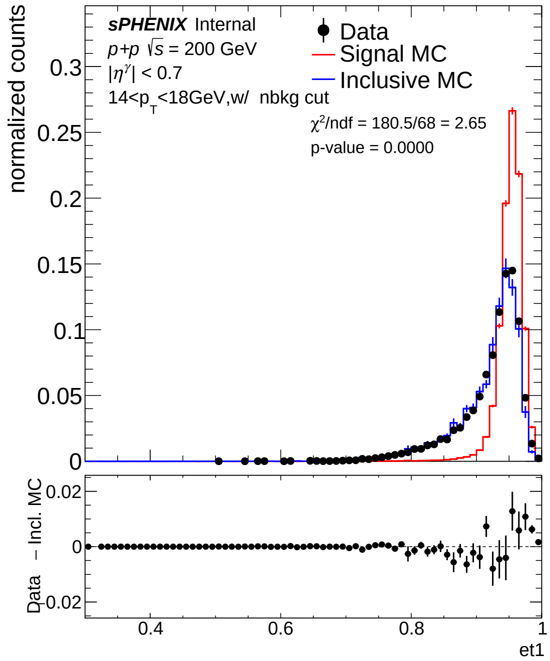


(a) Nominal MC

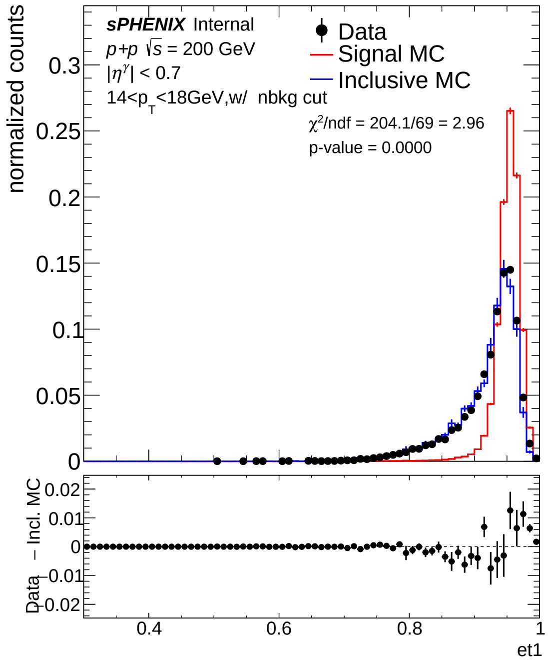
(b) Mixed MC (7.9% DI)

Figure 16: BDT score, $22 < E_{\mathrm{T}} < 28\,\mathrm{GeV}$ (pt3), 1.5 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 1.89$; mixed $\chi^2/\mathrm{ndf} = 1.13$ ($p = 0.249$; $1.7\times$ improvement).

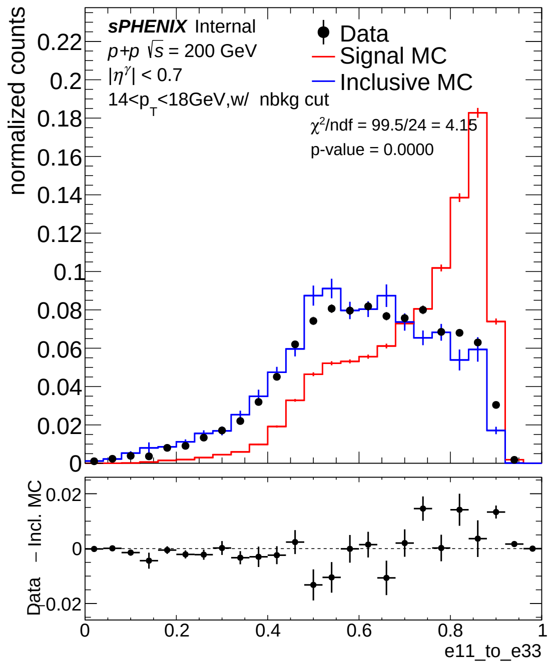
6.4.3 f_1 and $E_{1\times1}/E_{3\times3}$



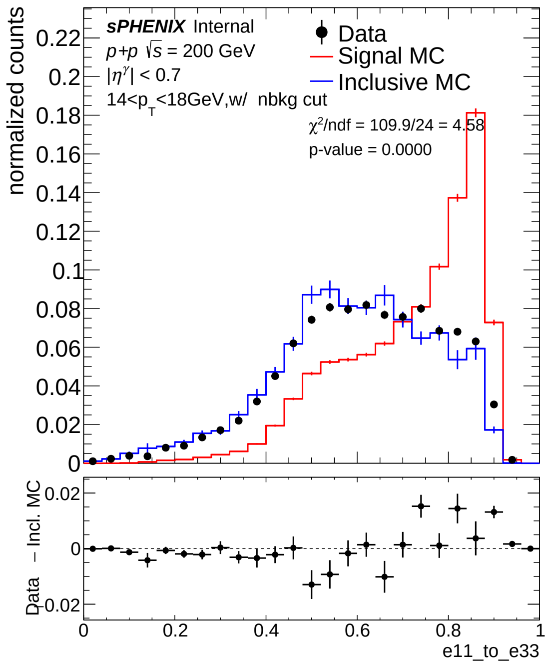
(a) f_1, nominal MC



(b) f_1, mixed MC (7.9% DI)



(c) $E_{1\times1}/E_{3\times3}$, nominal MC

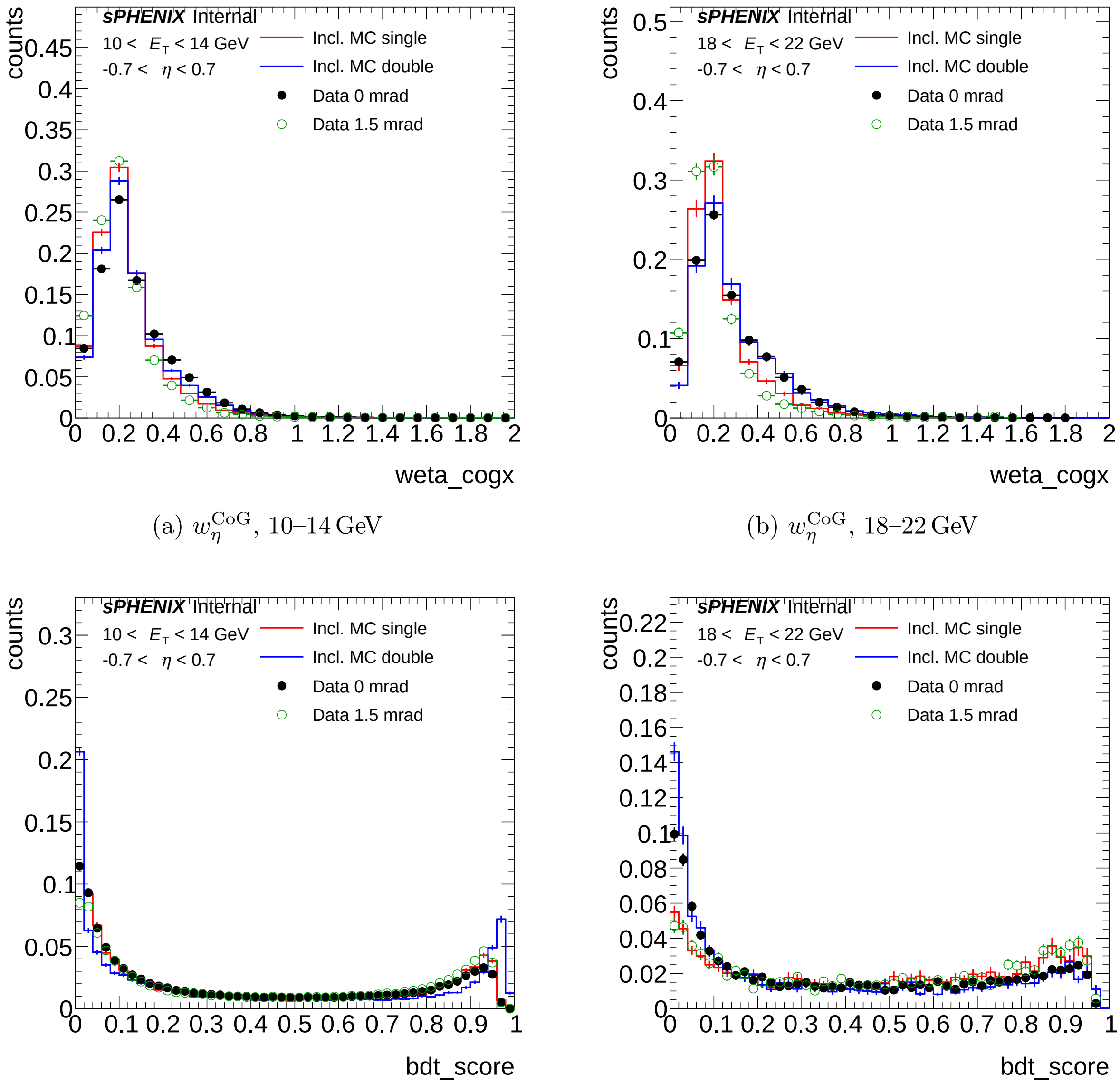


(d) $E_{1\times1}/E_{3\times3}$, mixed MC (7.9% DI)

Figure 17: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_{\mathrm{T}} < 18$ GeV (pt1), 1.5 mrad, cut1. f_1: $\chi^2/\mathrm{ndf} = 4.17 \to 2.55$ (1.6×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\mathrm{ndf} = 6.88 \to 6.04$ (1.1×).

6.5 Single-vs-Double Overlay with Data from Both Periods

Figures 18 and 19 overlay pure single-interaction MC (red) with pure double-interaction MC (blue) and data from both periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized at cut1. Data at 0 mrad systematically tracks the double-MC tail while data at 1.5 mrad tracks closer to single MC, consistent with the 22.4% vs. 7.9% DI fractions and providing a direct sample-level cross-check on the blending pipeline. The 8-panel overlay is split into two 4-panel figures for readability: the vertex-shift-sensitive observables (w_η^{CoG} and the composite BDT score) in Fig. 18, and the ring-energy observables (f_1 and $E_{1\times1}/E_{3\times3}$) in Fig. 19.



(a) w_η^{CoG}, 10–14 GeV

(b) w_η^{CoG}, 18–22 GeV

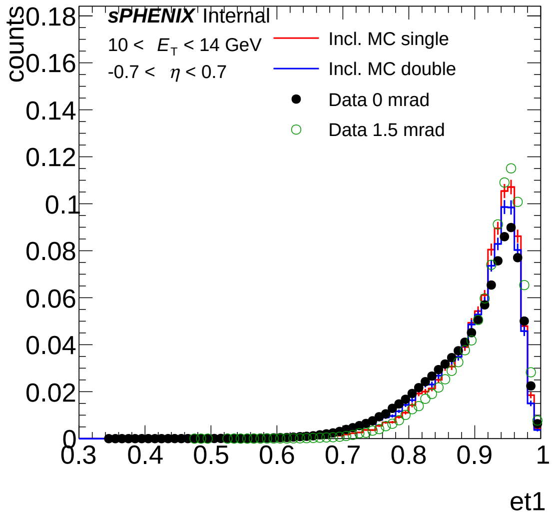
(c) BDT score, 10–14 GeV

(d) BDT score, 18–22 GeV

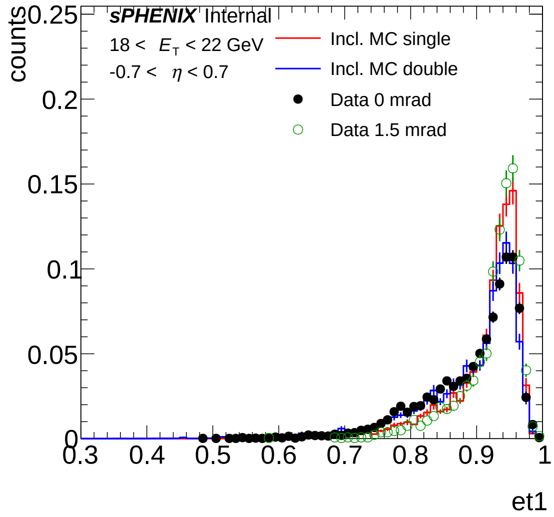
Figure 18: Sample-level shower-shape overlay, vertex-shift-sensitive observables: single-interaction MC (red), double-interaction MC (blue), data 0 mrad (filled black circles), data 1.5 mrad (open green circles), all area-normalized. Left column: $10 < E_{\mathrm{T}} < 14$ GeV. Right column: $18 < E_{\mathrm{T}} < 22$ GeV. Top row: w_η^{CoG}. Bottom row: BDT score. 0 mrad data systematically tracks the DI tail; 1.5 mrad data tracks closer to single MC.

6.6 χ^2/ndf Summary

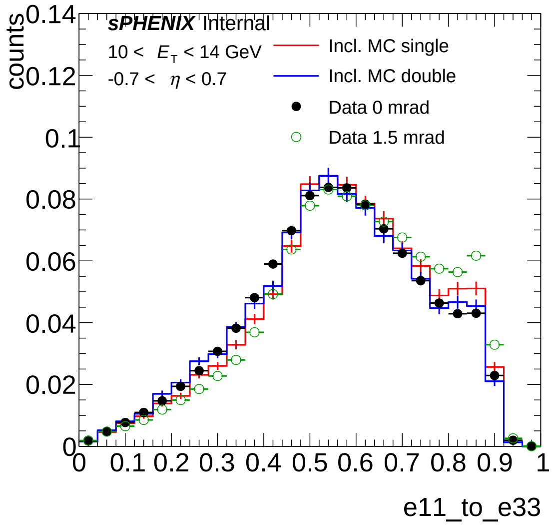
Table 8 compiles the Data − Inclusive-MC χ^2/ndf and p-values for all variables, E_{T} bins, and crossing angles. Entries where the mixed MC *degrades* agreement relative to nominal are high-



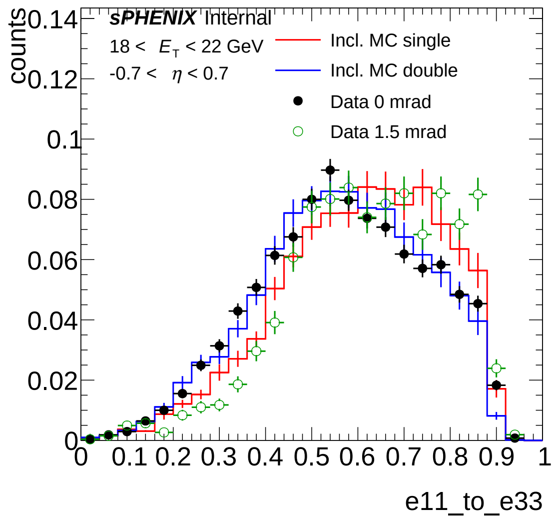
(a) f_1, 10–14 GeV



(b) f_1, 18–22 GeV



(c) $E_{1\times1}/E_{3\times3}$, 10–14 GeV



(d) $E_{1\times1}/E_{3\times3}$, 18–22 GeV

Figure 19: Sample-level shower-shape overlay, ring-energy observables: colour coding as in Fig. 18. Left column: $10 < E_{\mathrm{T}} < 14$ GeV. Right column: $18 < E_{\mathrm{T}} < 22$ GeV. Top row: f_1. Bottom row: $E_{1\times1}/E_{3\times3}$. Both observables show smaller DI-induced single/double separation than w_η^{CoG}/BDT, and the DI-sensitivity ordering between periods still holds.

lighted in red.

Table 8: Data – Inclusive-MC χ^2/ndf and p-values at cut1 for nominal (single-only) and mixed (double-blended) MC at both crossing angles. Lower χ^2/ndf is better; $p \ll 1$ indicates significant data–MC disagreement. Red entries flag cases where the blend degrades agreement.

		0 mrad (22.4% DI)				**1.5 mrad (7.9% DI)**			
Variable	E_T [GeV]	Nom.	p	Mixed	p	Nom.	p	Mixed	p
w_η^{CoG}	10–14 (pt0)	47.40	<0.001	28.96	<0.001	9.72	<0.001	42.31	<0.001
w_η^{CoG}	14–18 (pt1)	17.64	<0.001	2.99	<0.001	6.73	<0.001	4.86	<0.001
w_η^{CoG}	22–28 (pt3)	15.58	<0.001	0.75	0.840	21.10	<0.001	1.31	0.118
BDT score	10–14 (pt0)	6.85	<0.001	20.30	<0.001	5.98	<0.001	33.08	<0.001
BDT score	14–18 (pt1)	12.20	<0.001	1.63	0.004	1.16	0.207	2.31	<0.001
BDT score	22–28 (pt3)	9.62	<0.001	1.75	0.001	1.89	<0.001	1.13	0.249
f_1	14–18 (pt1)	9.33	<0.001	1.92	<0.001	4.17	<0.001	2.55	<0.001
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.50	<0.001	4.14	<0.001	6.88	<0.001	6.04	<0.001

6.7 Low-E_{T} Crossover and Recommendation

A consistent pattern emerges across *both* crossing angles: the DI blending **degrades** data–MC agreement for the BDT score at low E_{T} (pt0, 10–14 GeV) while **improving** it at higher E_{T}. This is now confirmed for 0 mrad (3.0× BDT degradation at pt0) and 1.5 mrad (5.5× degradation at pt0). For w_η^{CoG}, the crossover lies between pt0 and pt1: at 0 mrad, pt0 w_η^{CoG} still improves (1.6×); at 1.5 mrad it degrades (4.4×). The crossover pileup fraction for w_η^{CoG} at low E_{T} therefore lies between 7.9% and 22.4%.

Three contributing mechanisms are plausible:

1. **Shared MC samples**: the same `photon10_double` and `jet12_double` samples are used for both crossing angles; the beam conditions in the MC may better match one period than the other, and the mismatch only becomes visible at low E_{T} where the shower-shape distributions are narrow.
2. **Low-E_{T} sensitivity**: clusters near the 8–14 GeV threshold are most sensitive to absolute vertex shifts, since a fixed Δz produces a larger fractional kinematic offset.
3. **Small-fraction amplification at 1.5 mrad**: at 7.9% mixing, any residual mismatch between the DI MC and the real pileup is amplified relative to the small physical correction.

These observations motivate the use of mixed MC as a shower-shape *validation* rather than as a shower-shape template correction, and the recommendation: apply mixed MC as a cross-check at $E_{\mathrm{T}} \geq 14$ GeV (0 mrad) or $E_{\mathrm{T}} \geq 18$ GeV (1.5 mrad), and take the nominal-mixed spread as a systematic in the lower bins. The residual shower-shape mismodelling at low E_{T} is expected to be subdominant after the truth-vertex reweight is applied (§7), which shares data with the shower-shape pipeline and whose downstream application carries the same low-E_{T} sensitivity.

7 Truth-Vertex Reweighting

The shower-shape and efficiency analyses use a single-variable vertex reweight to close the MC luminous-region shape to data. The production choice is the reco-level ratio $f(z_r) = D(z_r)/M_{\text{reco}}(z_r)$, which closes the reco-vertex distribution exactly. This section shows that $f(z_r)$ is *structurally insufficient* for double-interaction events, quantifies the resulting cross-section bias at 1.5 mrad, and presents the nominal fix: an iterative data-driven truth-vertex reweight $w(z)$, factorized as $w(z_h)$ for single MC and $w(z_h)\, w(z_{\text{mb}})$ for double MC.

7.1 Why the Production $f(z_r)$ Cannot Close the Truth Vertex

The efficiency denominator in `RecoEffCalculator_TTreeReader.C` applies a reco-vertex cut $|z_r| <$ `vertex_cut` *and* a truth-vertex cut $|z_h| <$ `vertex_cut` (both set to 60 cm in `config_bdt_nom.yaml`). The reco cut is handled correctly by $f(z_r)$: the reweighted MC reco distribution matches data bin-by-bin by construction. The truth cut is *not* handled correctly for double-interaction events.

Decompose the pair of truth vertices $(z_{t,1}, z_{t,2}) \equiv (z_h, z_{\rm mb})$ of a double-interaction event into a symmetric coordinate $\bar{z} \equiv (z_h + z_{\rm mb})/2$ and an antisymmetric coordinate $\varepsilon \equiv (z_h - z_{\rm mb})/2$. Under the independent Gaussian sampling of the raw MC, $\bar{z}$ and ε are statistically independent zero-mean Gaussians with $\sigma_{\bar{z}} = \sigma_\varepsilon = \sigma_{\rm MC}/\sqrt{2}$. The reco vertex is $z_r \approx \bar{z}$ (up to MBD resolution), so $f(z_r)$ depends *only* on $\bar{z}$ and is flat in ε. After reweighting, $\bar{z}$ matches the target Gaussian $\mathcal{N}(0, \sigma^2_{\rm lum}/2)$ by construction, but ε is unchanged. The primary truth vertex $z_h = \bar{z} + \varepsilon$ is therefore a sum of a narrowed and an un-narrowed Gaussian, with variance

$$\sigma^2(z_h) = \tfrac{1}{2}\,\sigma^2_{\rm lum} + \tfrac{1}{2}\,\sigma^2_{\rm MC} = \tfrac{1}{2}(22.3^2 + 62.7^2) \simeq (47.0\,{\rm cm})^2. \tag{6}$$

We use $\sigma_{\rm lum} \simeq \sigma_{\rm data\,reco}$ here (22.3 cm rather than the 21.48 cm $\sigma_{\rm seed}$ quoted in §7.3); MBD resolution is sub-cm, so the two widths agree to within 1 cm at 1.5 mrad and the narrow-Gaussian approximation to the luminous region is good at the required precision. Any principled fix must act on the truth vertices directly, i.e. factorize as $w(z_h)\,w(z_{\rm mb})$ rather than depending only on the reco combination z_r.

7.2 Cross-Section Impact at 1.5 mrad

For a Gaussian truth distribution of width σ, the fraction inside $|z_h| < 60$ cm is ${\rm erf}(60/(\sigma\sqrt{2}))$:

- $\sigma = 47$ cm (what nominal $f(z_r)$ gives for double MC): ${\rm erf}(60/(47\sqrt{2})) \simeq 0.80$
- $\sigma = 22$ cm (target for 1.5 mrad): ${\rm erf}(60/(22\sqrt{2})) \simeq 0.99$

Double-interaction MC therefore loses $\sim$19 percentage points of truth fiducial acceptance that it should not. At 1.5 mrad the mixed blend is $0.921 \times {\rm single} + 0.079 \times {\rm double}$, so the bias on the integrated efficiency denominator is

$$0.079 \times 0.19 \simeq 1.4\,\% \quad \text{at 1.5 mrad.}$$

This translates directly to a +1.4% bias on the extracted cross section at 1.5 mrad (the efficiency denominator is too small, so the quoted cross section is too large). At 0 mrad the analogous bias is $\sim$0.3% because the raw MC beam profile already sits close to the wider 0 mrad luminous region. This is comparable to other PPG12 systematics and worth fixing.

7.3 Method: Data-Driven Iterative Truth-Vertex Reweight

The method derives a truth-vertex weight function $w(z)$ directly from the data reco distribution, with no closed-form Gaussian assumption on the luminous-region shape.

Forward model. Let f_d be the cluster-weighted double-interaction fraction (0.079 at 1.5 mrad, 0.224 at 0 mrad) and $f_s = 1 - f_d$. Under a candidate weight $w(z)$, the mixed-MC reco distribution is

$$M(z_r) = f_s \sum_{e \in T_s} w(z_h^e)\,\delta(z_r - z_r^e) + f_d \sum_{e \in T_d} w(z_h^e)\,w(z_{\rm mb}^e)\,\delta(z_r - z_r^e), \tag{7}$$

evaluated event-by-event over the single-interaction sample T_s and the double-interaction sample T_d. For double-interaction events the weight enters *twice*, once per truth vertex. This factorization gives the iteration access to the antisymmetric coordinate ε of §7.1 that the production reco-only weight $f(z_r)$ cannot touch.

Binning and closure window. Both periods use 80 bins over $[-200, 200]$ cm (bin width 5 cm), with identical bin edges for $w(z)$ and for the reco-vertex histograms. The closure window is auto-derived per period as the 99-th percentile of $|z_r^{\rm data}|$: 1.5 mrad uses $z_{\rm cut} = 83$ cm (34 bins), 0 mrad uses $z_{\rm cut} = 144$ cm (58 bins). Outside the window, data statistics are too low for the iteration to constrain $w(z)$.

Seed. $w_0(z) = \mathcal{N}(0, \sigma^2_{\rm seed})/\mathcal{N}(0, \sigma^2_{\rm gen})$, with $\sigma_{\rm gen} = 64.997 \pm 0.015$ cm measured from the photon10 primary truth vertex (identical between periods, since the same MC is used for both), and $\sigma_{\rm seed} = {\rm std}(z_r^{\rm data})$ measured from data — 21.48 cm at 1.5 mrad, 54.32 cm at 0 mrad. $\sigma_{\rm seed}$ requires *no* physics input: no Piwinski narrowing factor, no luminous-region calculation; it is just the empirical width of the reco distribution we are trying to match.

Iterative update. At iteration i the residual is $r_i(z_r) = D(z_r)/M_i(z_r)$. The update is Lucy–Richardson-like, multiplicative, and damped:

$$w_{i+1}(z) = w_i(z) \cdot \left[\text{smooth}(r_i(z))\right]^\alpha, \qquad \alpha = 0.5, \tag{8}$$

where the residual at reco bin z_r is applied to the weight at the matching truth bin $z = z_r$. After each update w is renormalized so $\langle w(z_h) \rangle = 1$ averaged over the single-MC truth distribution (preserving the physics DI fraction). Damping prevents over-correction; 1-bin Gaussian smoothing on the residual prevents bin-by-bin oscillation; the update factor is clipped to $[0.5, 2]$ per iteration; boundary bins are edge-clamped. The final $w(z)$ is data-driven everywhere, without a post-iteration snap to a seed Gaussian (an earlier variant did snap; `compare_snap_strategies.py` showed removing the snap improves 0 mrad χ^2 by $\sim$1.3× and max $|r - 1|$ by $\sim$3× while leaving 1.5 mrad essentially unchanged).

Convergence is $\max |r_i(z) - 1| < 0.01$ inside the closure window, with a hard cap of 30 iterations.

Implementation. Standalone Python tool in `efficiencytool/truth_vertex_reweight/`. The mixed-MC histogram is heavy to rebuild ($\sim$22 M events at 1.5 mrad, $\sim$48 M at 0 mrad), so per-sample arrays are cached to `output/cache/*.npz` keyed on file mtime. Each iteration runs in seconds per iteration on a single core. Outputs live at `output/{1p5mrad,0mrad}/reweight.root` (containing `h_w_iterative`, `h_w_seed`, and `h_M_iterative`) and `fit_result.yaml`. The code is driven from the period configs `config_showershape_{1p5rad,0rad}.yaml` via the new `truth_vertex_reweight` and `truth_vertex_reweight_file` fields.

7.4 Per-Period Results

Both periods are fit on the photon10/photon10_double mix only at this stage; the jet12 extension is a follow-up (§7.7). Table 9 collects the headline numbers.

quantity	1.5 mrad	0 mrad
run range	51274–54000	47289–51274
luminosity [pb^{-1}]	16.8588	32.6574
f_{double} (cluster-weighted)	0.079	0.224
data $\lvert z_r \rvert$ 99% percentile [cm]	82.0	143.9
closure window z_{cut} [cm]	83	144
bins inside closure window	34	58
$\sigma_{\mathrm{seed}} = \mathrm{std}(z_r^{\mathrm{data}})$ [cm]	21.48	54.32
σ_{gen} from MC z_h (generator) [cm]	64.997 ± 0.015	
initial χ^2 (in window)	1,419,454	26,589
final χ^2 (in window)	453	158
χ^2 reduction factor	3132×	168×
final χ^2/ndf (in window)	13.7 (ndf 33)	2.78 (ndf 57)
initial max $\lvert r-1 \rvert$	0.635	0.079
final max $\lvert r-1 \rvert$ (in window)	0.111	0.008
iterations used	30 (cap hit)	8
converged to 0.01 tolerance	**no**	**yes**
(context) parametric Gaussian χ^2/ndf	3237	454

Table 9: Iterative data-driven truth-vertex reweight, photon10 mix only, with the post-iteration snap to seed removed. The 0 mrad fit converges to the 1% per-bin tolerance in 8 iterations (final max $\lvert r-1 \rvert \simeq 0.8\%$). The 1.5 mrad fit reduces in-window χ^2 by a factor $\sim$3100 in 30 iterations and stalls at max $\lvert r-1 \rvert \simeq 11\%$, dominated by bins at the closure-window edge $\lvert z_r \rvert \simeq 83$ cm. Inside the bulk $\lvert z_r \rvert < 30$ cm the per-bin closure is at the 1–2% level. The last row reports the parametric best-fit Gaussian χ^2/ndf from a 1-parameter σ_{tgt} scan; the iterative method improves on this by a factor $\sim$236 at 1.5 mrad and $\sim$163 at 0 mrad.

Hero figure. Figure 20 is the single-figure summary of the iterative method across both periods: data reco vs. reweighted MC reco, residuals, the recovered $w(z)$, and the headline numbers.

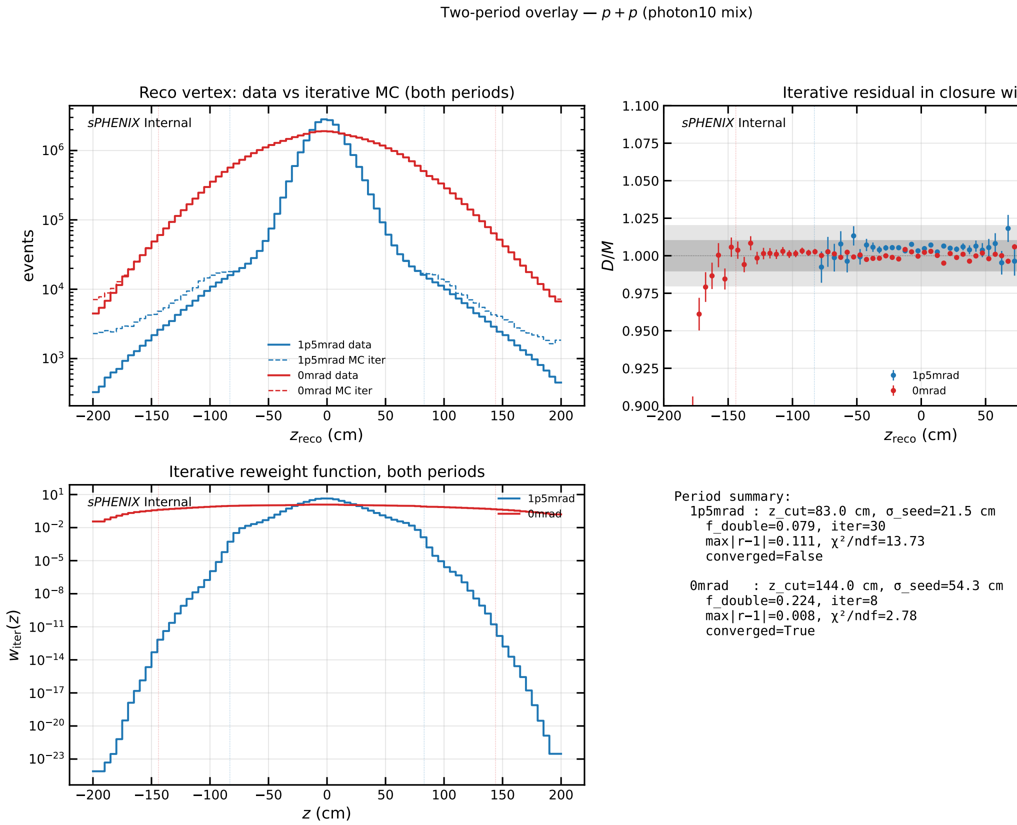


Figure 20: Iterative data-driven truth-vertex reweight summary, both crossing angles. Top-left: data reco z_r (solid) vs. mixed MC reco z_r after iterative reweight (dashed), both periods overlaid (log scale). Top-right: residual $r(z_r) = D(z_r)/M_{\text{iter}}(z_r)$ inside each period's closure window (vertical dashed lines). Bottom-left: recovered truth-vertex weight $w(z)$ for both periods (log scale). Bottom-right: summary statistics. The 0 mrad case is essentially closed; the 1.5 mrad case is closed in the bulk and stalls at the closure-window edge.

Truth vertex comparison across methods. Figure 21 overlays the primary truth vertex z_h distribution of the mixed MC under three weighting methods: no reweight (raw MC, $\sigma \simeq 64$ cm), the production reco-level weight $f(z_r)$, and the new iterative truth-level weight $w(z_h)$. At 1.5 mrad the production method leaves z_h at $\sigma \simeq 27$ cm — narrower than the raw MC but still far wider than the true luminous-region width ($\sigma_{\text{lum}} \simeq 22$ cm). The iterative method narrows z_h to $\sigma \simeq 16$ cm, which is narrower than the data reco width $\sigma_{\text{seed}} = 21.5$ cm precisely as expected (the MBD vertex finder adds resolution smearing, so each truth vertex must be narrower than the reco width). At 0 mrad both methods converge at $\sigma \simeq 54$–57 cm.

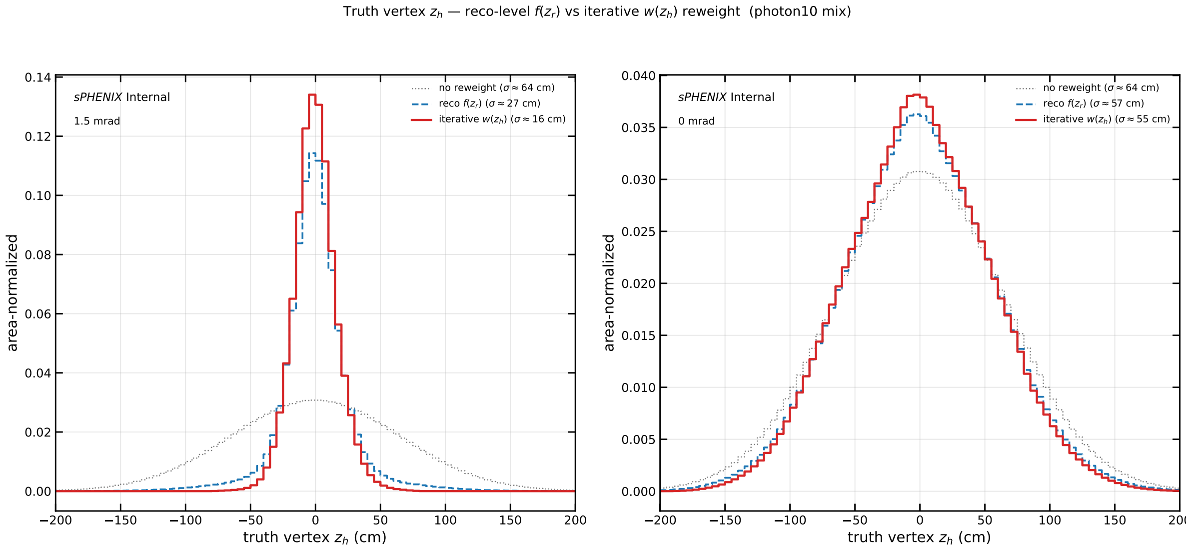


Figure 21: Truth vertex z_h distribution of the mixed photon10 MC under three reweighting methods, per period. Left: 1.5 mrad, where $f(z_r)$ (blue dashed) fails to narrow the truth distribution sufficiently while the iterative $w(z_h)$ (red) narrows it past the data reco width. Right: 0 mrad, where both methods agree because the MC beam profile is already close to the data luminous region.

0 mrad: clean convergence. Figure 22 shows the 6-panel closure diagnostic for 0 mrad. The seed Gaussian with $\sigma_{\text{seed}} \simeq 54$ cm already captures most of the shape (initial max $|r-1| = 8\%$, initial $\chi^2/\text{ndf} \simeq 467$); the parametric best-fit Gaussian at $\sigma_{\text{tgt}} \simeq 54.6$ cm is essentially identical ($\chi^2/\text{ndf} \simeq 454$) because the data is nearly Gaussian. The iteration cleans up the residual over 8 updates and drops below the 1%-per-bin tolerance, with final max $|r-1| \simeq 0.8\%$ and $\chi^2/\text{ndf} \simeq 2.78$. The recovered $w(z)$ is smooth and broadly Gaussian with a small central enhancement.

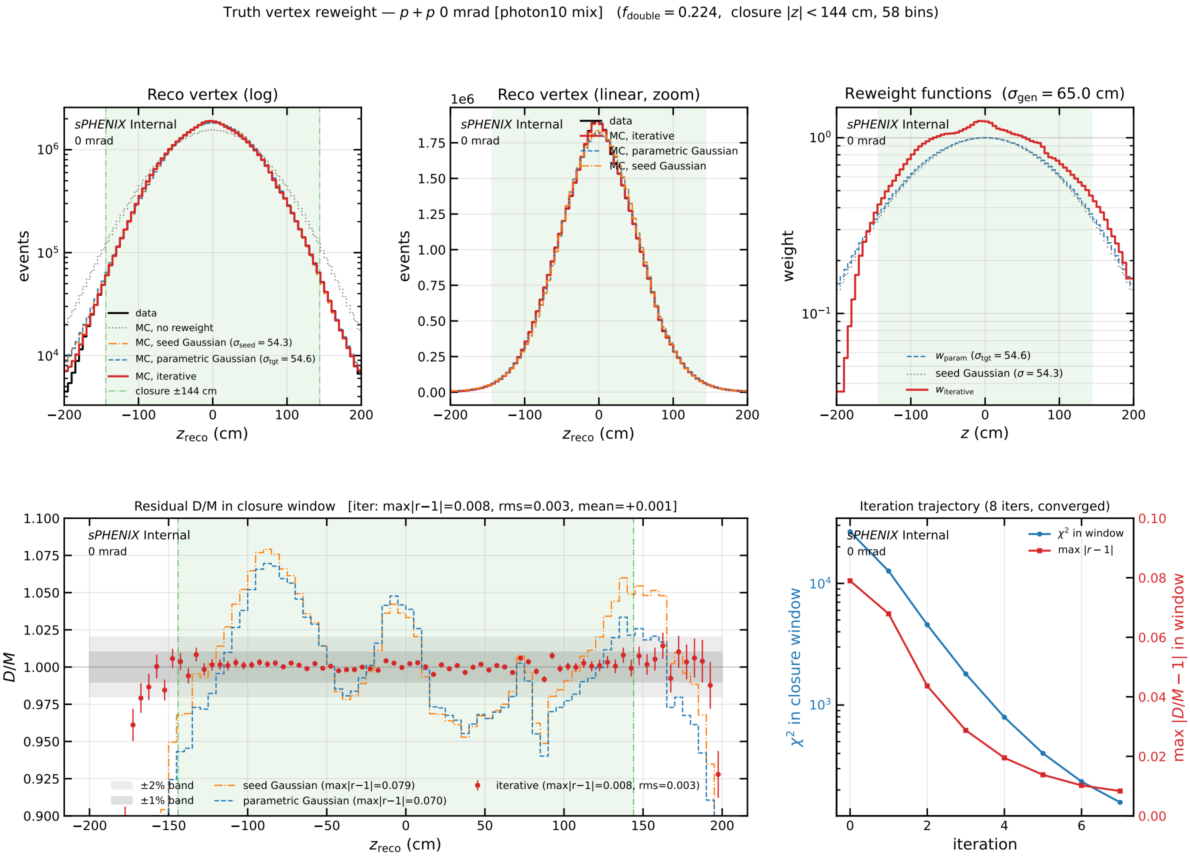


Figure 22: 0 mrad iterative closure (6 panels). Top-left/centre: data reco vs. mixed MC reco under no reweight (gray dotted), the seed Gaussian $w(z) \propto \mathcal{N}(0, \sigma^2_{\text{seed}})/\mathcal{N}(0, \sigma^2_{\text{gen}})$ (orange dash-dot), the parametric best-fit Gaussian (blue dashed), and the iterative reweight (red). Top-right: the corresponding $w(z)$. Bottom-left: residual D/M in the closure window with Poisson errors and $\pm 1\%/\pm 2\%$ reference bands. Bottom-right: per-iteration χ^2 and max $|r-1|$ trajectory. At 0 mrad the seed and the parametric Gaussian are nearly identical because the data reco is already close to a single Gaussian; the iterative method improves on both.

1.5 mrad: bulk closure, edge stall. Figure 23 shows the 6-panel diagnostic for 1.5 mrad. The seed Gaussian with $\sigma_{\text{seed}} \simeq 21$ cm starts from max $|r-1| = 64\%$ and an in-window χ^2 of 1.42×10^6. The first ~ 5 iterations make dramatic progress; the iteration then enters a slow plateau driven by bins at the closure-window edge $|z_r| \simeq 83$ cm. After 30 iterations χ^2 in the window has dropped by $\sim 3100\times$ to $\simeq 453$, and max $|r-1|$ sits at $\simeq 0.11$, set by those stuck edge bins. Inside the bulk $|z_r| < 30$ cm the closure is $\sim$1–2% per bin; between 30 and 60 cm the residual climbs to $\sim$5–10%. The recovered $w(z)$ shows a strong central core boost and a clear non-Gaussian shape in $|z| \simeq 40$–60 cm that no symmetric Gaussian ansatz can represent.

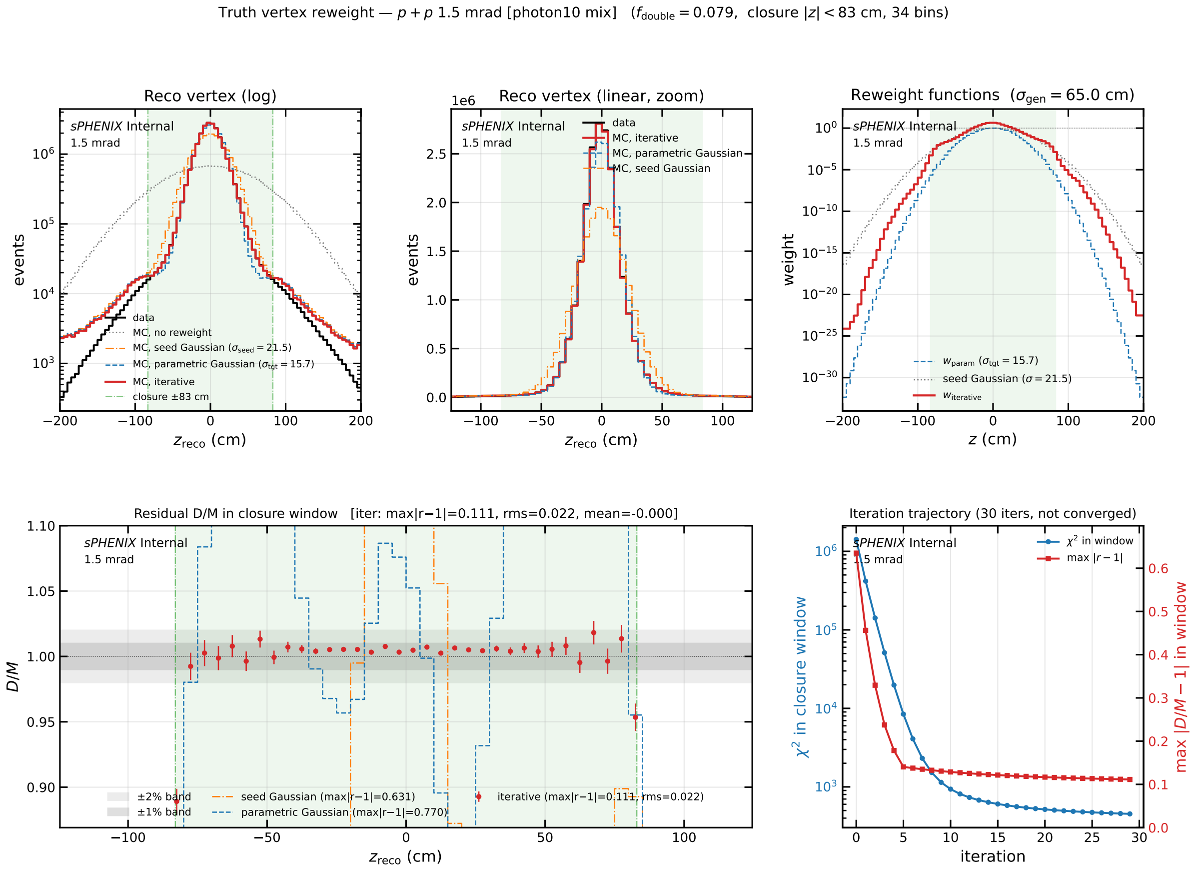


Figure 23: 1.5 mrad iterative closure (6 panels), same layout as Fig. 22. At 1.5 mrad the seed Gaussian ($\sigma_{\mathrm{seed}} = 21.5$ cm, orange dash-dot) and the parametric best-fit Gaussian ($\sigma_{\mathrm{tgt}} = 15.7$ cm, blue dashed) are distinct; both mis-predict the reco-vertex shape by 30–60% at the closure-window edge. The iterative method (red points) closes the bulk to 1–2% inside $|z_r| < 30$ cm and stalls at $\max|r-1| \simeq 0.11$ at $|z_r| \simeq 83$ cm. The recovered $w(z)$ in the top-right panel shows a strong central core boost and a clear non-Gaussian shape in $|z| \simeq 40$–60 cm.

Method comparison. Figure 24 overlays the iterative reweight against three reference baselines at 1.5 mrad: no reweight, the seed Gaussian (initial point of the iteration), and the parametric best-fit Gaussian (1-parameter σ_{tgt} scan with minimum at 15.7 cm). The no-reweight MC overshoots data by a factor of several; both Gaussian baselines match the width on average but mismatch the shape at 30–70%; only the iterative method follows the narrow data core *and* tracks the non-Gaussian mid-tail.

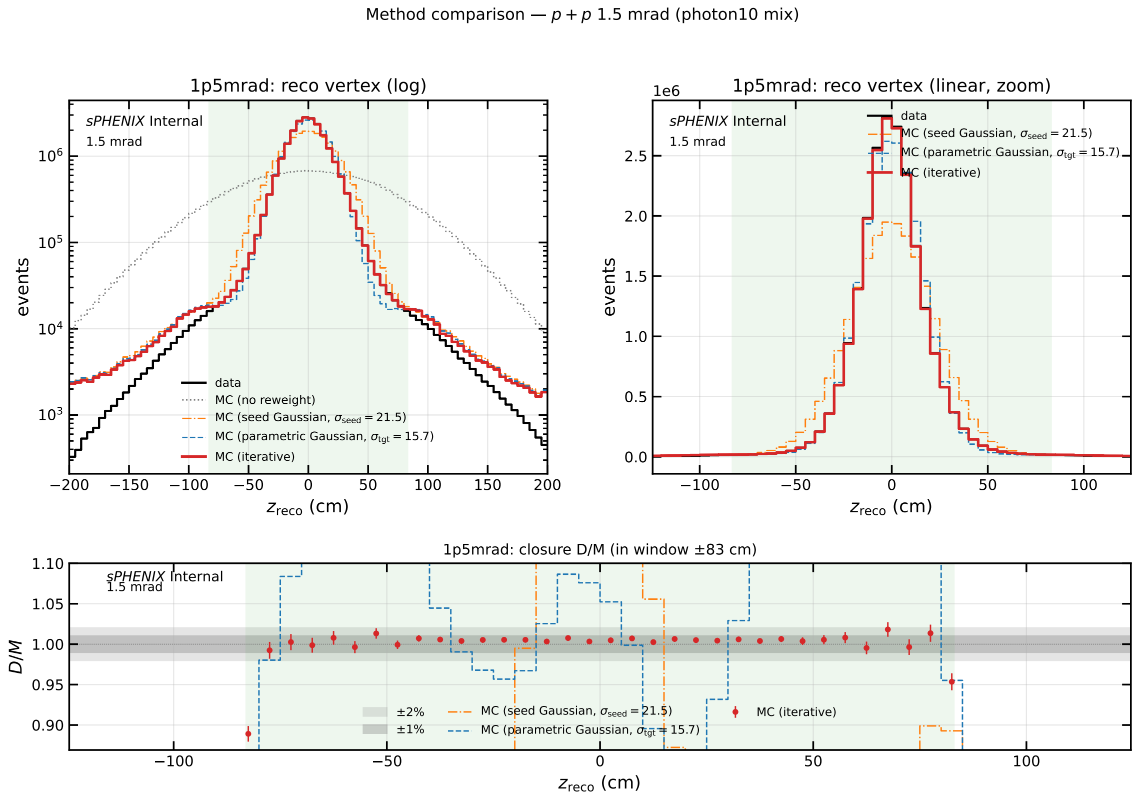


Figure 24: Method comparison at 1.5 mrad: data reco vs. mixed MC reco under no reweight (gray dotted), seed Gaussian $\sigma_{\rm seed} = 21.5$ cm (orange dash-dot), parametric best-fit Gaussian $\sigma_{\rm tgt} = 15.7$ cm (blue dashed), and the iterative data-driven reweight (red), with ratio panel below. The iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

Parametric $\sigma_{\rm tgt}$ scan. Figure 25 shows the stage-1 parametric χ^2 scan that precedes the iterative method. A single Gaussian truth-vertex weight $w(z) \propto \mathcal{N}(0, \sigma_{\rm tgt}^2)/\mathcal{N}(0, \sigma_{\rm gen}^2)$ is evaluated at each scan point. At 1.5 mrad the minimum is at $\sigma_{\rm tgt} \simeq 15.7$ cm with $\chi^2/\text{ndf} \simeq 3237$; the iterative method reaches 13.7, a $\sim 236\times$ improvement. At 0 mrad the minimum is at $\sigma_{\rm tgt} \simeq 54.6$ cm with $\chi^2/\text{ndf} \simeq 454$; the iterative method reaches 2.78, a $\sim 163\times$ improvement.

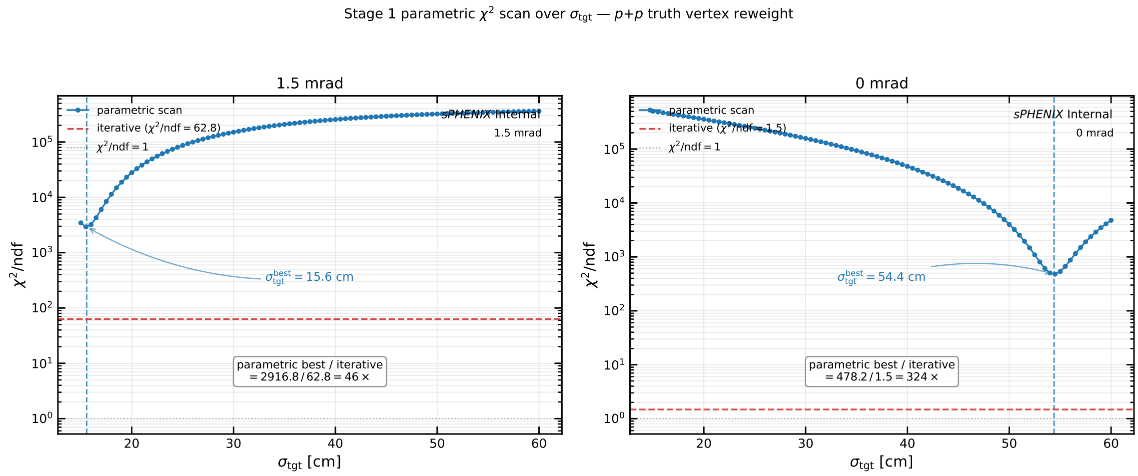
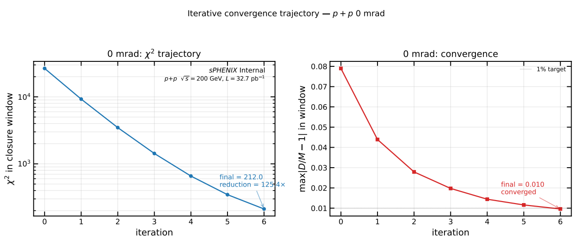
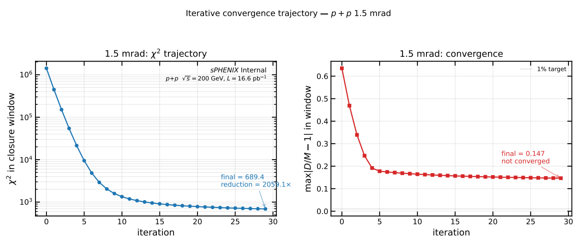


Figure 25: Stage-1 parametric χ^2/ndf as a function of the target truth-vertex Gaussian width σ_{tgt}, for the 1.5 mrad (left) and 0 mrad (right) periods. Vertical dashed line: parabolic-refined minimum. Horizontal dashed line (red): final χ^2/ndf achieved by the iterative method. Horizontal dotted line: $\chi^2/\text{ndf} = 1$. The gap between the parametric minimum and the iterative floor quantifies the non-Gaussian shape mismatch that no single-Gaussian ansatz can capture.

Convergence and pull diagnostics. Figures 26a–26b show the per-iteration trajectory of the χ^2 and $\max|r-1|$ metrics. Figures 27a–27b show the pull distribution of $D - M$ in the closure window.

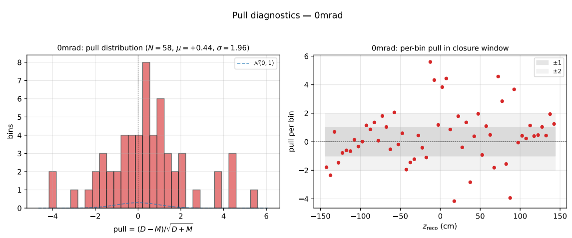


(a) 0 mrad convergence

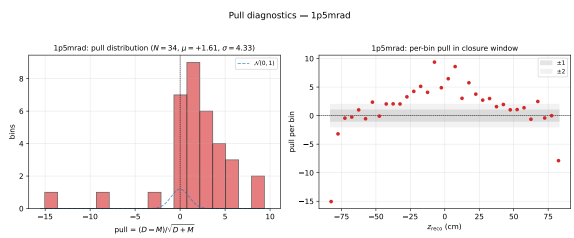


(b) 1.5 mrad convergence

Figure 26: Per-iteration χ^2 and $\max|r-1|$ trajectory for the iterative reweight. Left: 0 mrad (converged in 8 iterations). Right: 1.5 mrad (cap hit at 30 iterations, stalled at the closure-window edge).



(a) 0 mrad pull



(b) 1.5 mrad pull

Figure 27: Residual pull distributions in the closure window after the iterative reweight. 0 mrad (left) is centred and roughly unit-width; 1.5 mrad (right) shows a small tail at the closure-window edge.

7.5 Physics Interpretation

Why 0 mrad is easy. The 0 mrad luminous region ($\sigma_{\text{lum}} \simeq 54$ cm) is close to the generator beam profile ($\sigma_{\text{gen}} \simeq 65$ cm), and the data reco shape is nearly Gaussian. The seed is already a good first approximation and the iteration just cleans up small departures (slightly enhanced central core).

Why 1.5 mrad is hard. Three things conspire:

- The Piwinski narrowing compresses the luminous region to $\sigma_{\text{lum}} \simeq 22$ cm, nearly a factor of three narrower than the generator profile. The reweight must extract a narrow target from a wide source; the dynamic range of $w(z)$ at the central core is correspondingly large ($w(0) \simeq 8$).
- The data reco distribution is *not* a clean Gaussian. It has a narrow core plus a mid-tail shoulder near $|z_r| \simeq 45$–60 cm. The iterative method discovers this structure directly — the recovered $w(z)$ is non-Gaussian in the same z range — whereas a single-parameter Gaussian cannot represent it.
- The MBD resolution tail leak from core-truth events fills the bins near the closure-window edge ($|z_r| \simeq 83$ cm). The iteration becomes insensitive to $w(z)$ at that z because the contributions to those reco bins come dominantly from truth vertices at *other* z values via the wide double-interaction reco-truth kernel (Appendix A). This sets the residual floor at $\max |r - 1| \simeq 11\%$.

What the recovered $w(z)$ looks like. At 1.5 mrad the recovered $w(z)$ has three distinguishing features (top-right panel of Fig. 23): (a) a strong central peak with $w(0) \simeq 8$ that pulls doubles whose truth vertices both lie near the IP into the data-like core; (b) a mid-z dip that suppresses the generator's natural $|z| \sim 30$–50 cm population; and (c) a clear $\pm$ asymmetry in $|z| \simeq 40$–60 cm tracking the asymmetric mid-tail bump in the data. None of these features is representable by a symmetric single-parameter Gaussian. At 0 mrad, $w(z)$ is essentially the seed Gaussian with a small central enhancement.

7.6 Downstream Hookup

Both `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` now accept two analysis fields: `truth_vertex_reweight_on` (0/1) and `truth_vertex_reweight_file` (path to `reweight.root`). When `on=1`, the code reads `h_w_iterative`, multiplies the per-event weight by $w(z_h)$ for single-interaction MC and by $w(z_h)\, w(z_{\text{mb}})$ for double-interaction MC. The secondary truth vertex branch is picked up through an optional-pointer guard, so single-MC trees without the branch fall through to the single-vertex path. The reco-level and truth-level reweights are mutually exclusive; the config loader forces `vertex_reweight_on=0` when `truth_vertex_reweight_on=1`. The showershape configs `config_showershape_{1p5rad,0rad}.yaml` are pre-wired with the matching `reweight.root`; the nominal BDT configs carry the same fields but disabled, so flipping the main analysis over to the truth-vertex reweight is a one-line config edit once the jet-sample extension is done.

7.7 Caveats and Next Steps

- **photon10-only mix**: the current fit uses `photon10+photon10_double` only; the full production blend also includes `jet12+jet12_double`. Extending the fit is mechanically two more `load_sample` calls plus the appropriate cluster-weighted blend, but the closure numbers will move and must be remeasured before any production hookup.

- **1.5 mrad not formally converged**: the 30-iteration cap was hit. Open questions: (a) whether extending to ~ 100 iterations pushes the stuck edge bin to convergence or whether it is a hard physical floor set by the DI reco-truth kernel (the suspicion is the latter; Appendix A); (b) whether the failure to converge in $\max|r-1|$ matters downstream. The latter has a clean answer: the PPG12 analysis applies fiducial cuts $|z_r| < 60$ cm and $|z_h| < 60$ cm, inside which the per-bin closure is already 1–2%. The stuck edge at $|z_r| \simeq 83$ cm is well outside the analysis fiducial, so the iteration is good enough for the analysis use case.

- **Implementation can be sped up**: each iteration currently rebuilds the MC reco histogram from scratch ($\mathcal{O}(N_{\text{events}})$ per pass). Precomputing z^2 and dropping the per-trial mask rebuild would give a $\sim 10\times$ speedup with no change in result. Not a blocker; nice-to-have for the all-samples extension.

- **DI-MC `vertexz_truth_mb` branch**: the `photon10_double` and `jet12_double` `bdt_split.root` files currently fed to the efficiency pipeline predate the `vertexz_truth_mb` branch in the anatreemaker output, so $w(z_{\text{mb}})$ is silently dropped (optional-pointer guard). The impact is expected sub-percent at 1.5 mrad and sub-percent at 0 mrad; regenerating `bdt_split.root` from the reprocessed `combined.root` (which does carry the branch) removes this caveat.

8 Cluster Size Under DI

Cluster size (EMCal tower count n_{owned} plus the 7×7 bounding-box widths Δi_η, Δi_ϕ) is the basic shape property that underlies every shower-shape-based discriminator used in PPG12 (NPB, shower-shape BDT, width variables). This section reports the DI-induced shifts, split by sample, selection stage, and crossing-angle period.

8.1 Sample-Level Reference (Pre-Cut)

Figure 28 shows the pure single vs. pure double reference for photon10; Figure 29 for jet12. The mixed curves at 7.9% (1.5 mrad) and 22.4% (0 mrad) are bin-by-bin linear combinations of the two means, i.e. $\langle y\rangle_{\text{mix}}(E_{\text{T}}) = (1-f)\langle y\rangle_{\text{single}}(E_{\text{T}}) + f\langle y\rangle_{\text{double}}(E_{\text{T}})$, which avoids biasing the blend by the per-sample event counts.

Photons. DI shifts photon cluster size *upwards* by $\Delta\langle n_{\text{owned}}\rangle \approx +0.15$ to $+0.22$ towers above single across $E_{\text{T}} \in [15, 30]$ GeV— real but small. Pile-up adds a few soft neighbours to an otherwise compact EM shower. Bounding-box widths shift by $\lesssim 0.1$; the first-order effect on photon shower width is at the sub-0.1-index level. At 7.9% (1.5 mrad), the mixed curve is nearly indistinguishable from single — consistent with the shower-shape BDT being pileup-robust in the 1.5 mrad regime.

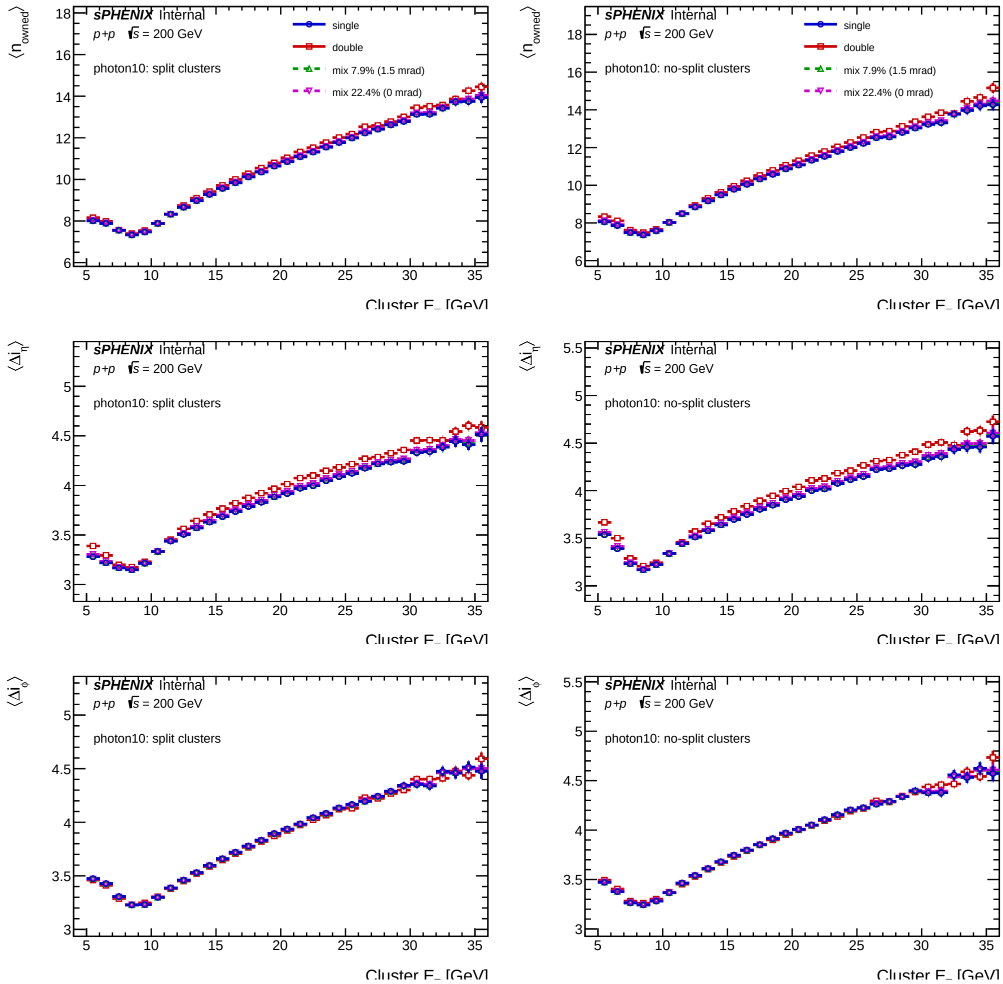


Figure 28: Photon DI reference: `photon10_nom` (blue) vs. `photon10_double` (red), plus mixed at $f = 7.9\%$ (green, 1.5 mrad) and $f = 22.4\%$ (magenta, 0 mrad). Pre-cut, both containers.

Jets. DI inflates *jet* cluster size much more strongly: $\Delta\langle n_{\text{owned}}\rangle \approx +0.5$ to $+1.3$ towers in the split container at $E_{\text{T}} > 15\,\text{GeV}$, roughly $4\times$ larger than the photon response. Physically: a jet cluster already has multiple soft neighbouring towers from accompanying hadrons, and DI adds more of the same; the probability that the DI overlay lands on a low-energy tower and pushes it over the 70 MeV threshold is much higher in a jet environment than for a clean photon. Because jets enter both the fake-photon and anti-tight control regions of the ABCD, this DI effect is a non-trivial input to the background-template closure: in 0 mrad (22.4% DI) running, the jet template mean cluster size shifts upward by ~ 0.3–0.4 towers, which affects the shower-shape BDT input distribution and therefore the background rejection rate.

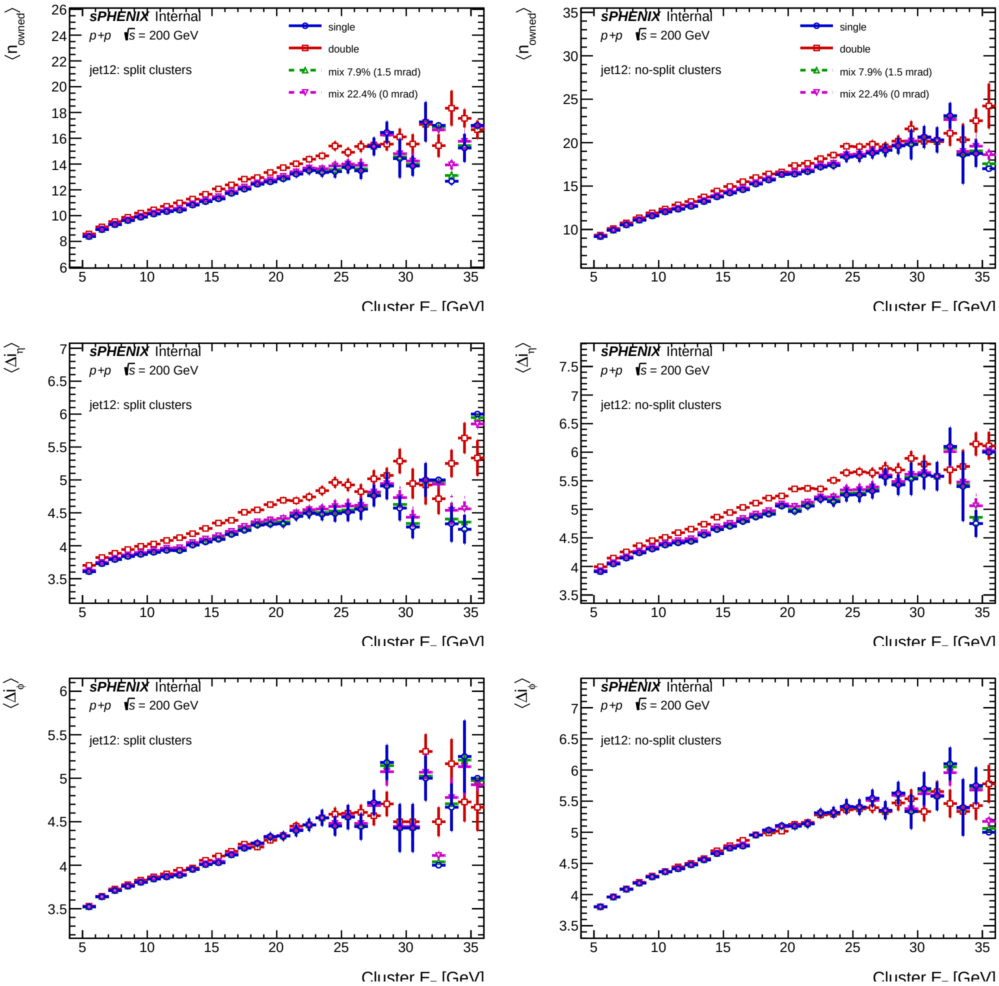


Figure 29: Jet DI reference: `jet12_nom` vs. `jet12_double` in the same format as Fig. 28. DI effect on jet cluster size is $\sim 4\times$ larger than on photon, consistent with softer neighbouring towers getting pushed above the 70 MeV threshold.

8.2 Selection-Stage Progression (With Truth-Vertex Reweight)

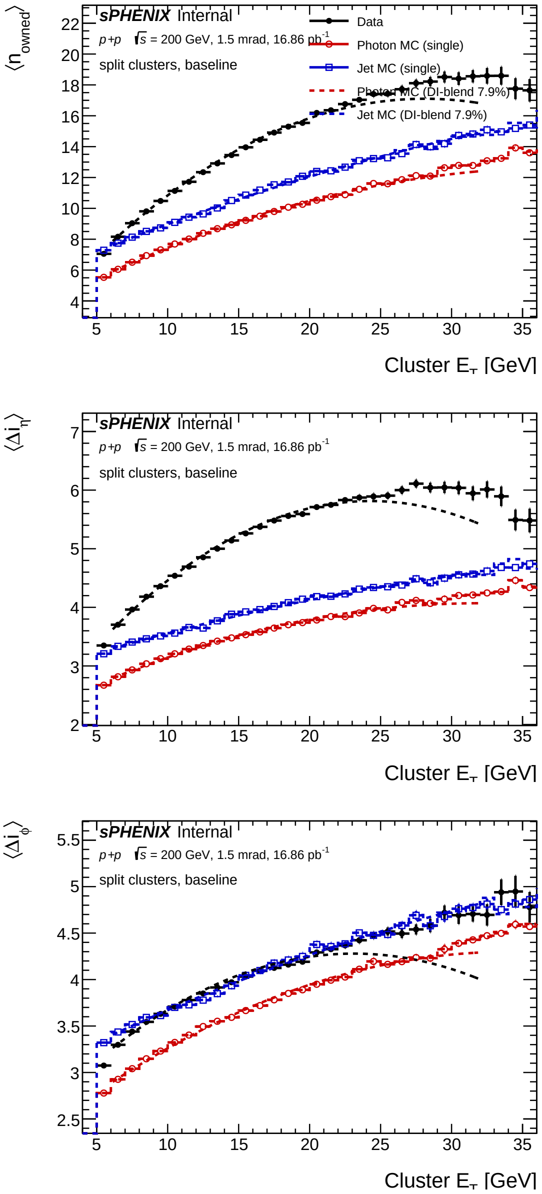
With the nominal iterative truth-vertex reweight applied, three selection stages are compared per period (baseline, common-cut, tight-cut), re-binned to the $E_{\mathrm{T}} \in [19, 21]$ GeV bin centre of the split container. The numerical values at $E_{\mathrm{T}} \simeq 20$ GeV are summarized in Table 10.

metric	sample	1.5 mrad baseline	1.5 mrad common	1.5 mrad tight	0 mrad baseline	0 mrad common	0 mrad tight
$\langle n_{\text{owned}} \rangle$	data	15.86	11.01	9.92	14.37	11.77	9.98
	photon	10.44	10.39	10.21	10.56	10.51	10.32
	jet	15.38	14.33	10.94	15.51	14.59	11.05
$\langle \Delta i_\eta \rangle$	data	5.65	3.90	3.63	5.10	4.15	3.66
	photon	3.78	3.77	3.72	3.83	3.82	3.76
	jet	4.81	4.63	3.84	4.85	4.69	3.88
$\langle \Delta i_\phi \rangle$	data	4.24	4.03	3.82	4.19	4.06	3.80
	photon	3.93	3.91	3.88	3.93	3.91	3.88
	jet	4.95	4.76	4.16	4.96	4.79	4.17

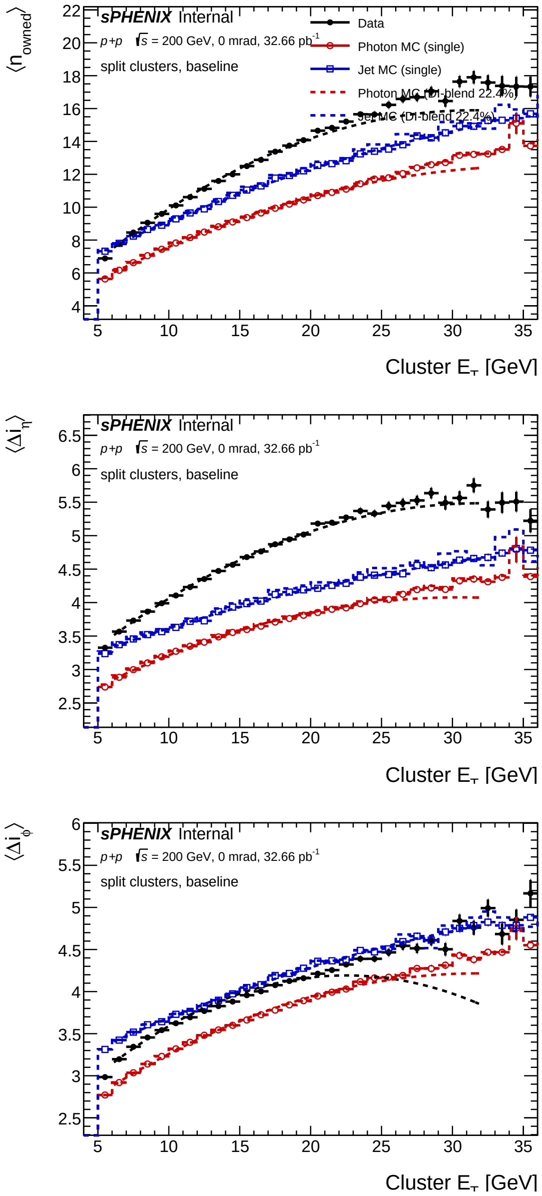
Table 10: Cluster-size means in the bin centre $E_T \in [19, 21]$ GeV, split container (`CLUSTERINFO_CEMC`), truth-vertex-reweighted MC. Values read from the per-period TProfiles in `results/cluster_size_cuts_PERIOD_SAMPLE.root` (PERIOD = 1p5mrad or 0mrad). Photon pool is photon5+photon10+photon20; jet pool is jet5+jet8+jet12+jet20+jet30+jet40. DI blending is applied in the figures only, not in this numeric table.

Three observations:

- **Common cut removes the noise-like tower excess, preferentially from data.** At 1.5 mrad and $E_T = 20$ GeV, $\langle n_{\text{owned}} \rangle$ drops from 15.86 → 11.01 in data (a 31% decrease), 15.38 → 14.33 in jet MC (only 7%), and 10.44 → 10.39 in photon MC (essentially unchanged). The common-cut's NPB+w_η-bound block targets noise/threshold contamination that is far more prevalent in data than in MC.
- **Tight cut converges data, jet MC, and photon MC to the signal cluster size.** At $E_T = 20$ GeV, 1.5 mrad: data tight 9.92, jet MC tight 10.94, photon MC tight 10.21 — a ±5% spread around a common ∼10.4-tower signal size. Photon-MC tight is the direct signal-MC comparator for data tight: the two agree to ∼3% on $\langle n_{\text{owned}} \rangle$ and $\langle \Delta i_\eta \rangle$ across $E_T \in [10, 30]$ GeV.
- **DI mixing is sub-leading in the tight selection.** At 1.5 mrad the DI-blend shift in $\langle n_{\text{owned}} \rangle$ is below 0.1 tower for every selection. At 0 mrad (22.4% DI) the jet-MC baseline shifts upward by ∼0.3 towers, but drops to below 0.1 tower after the tight cut. Photon-MC DI effect is below 0.1 tower at every stage and every period.

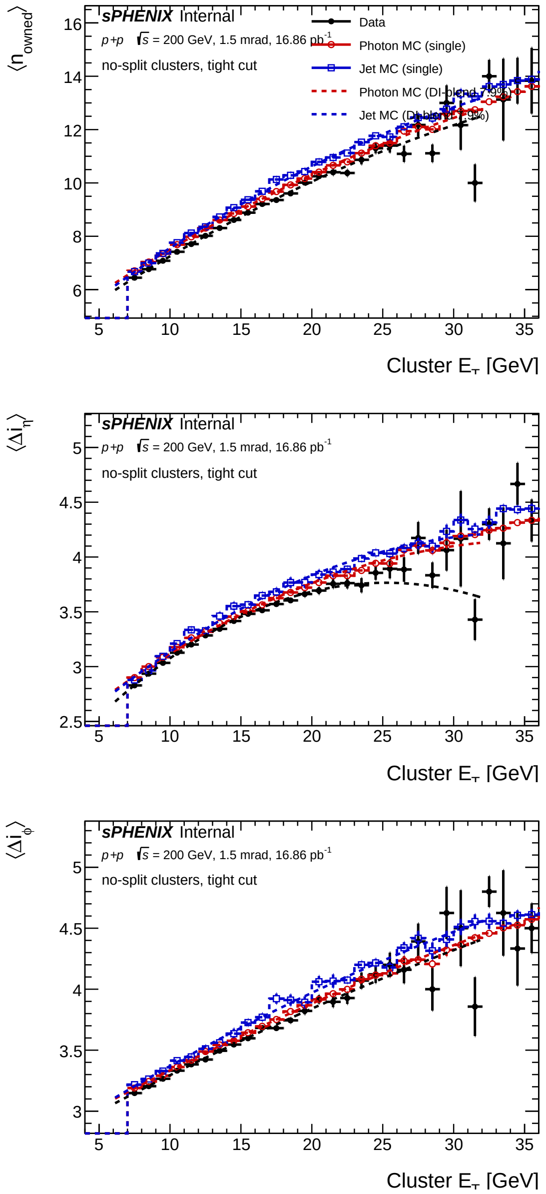


(a) 1.5 mrad baseline

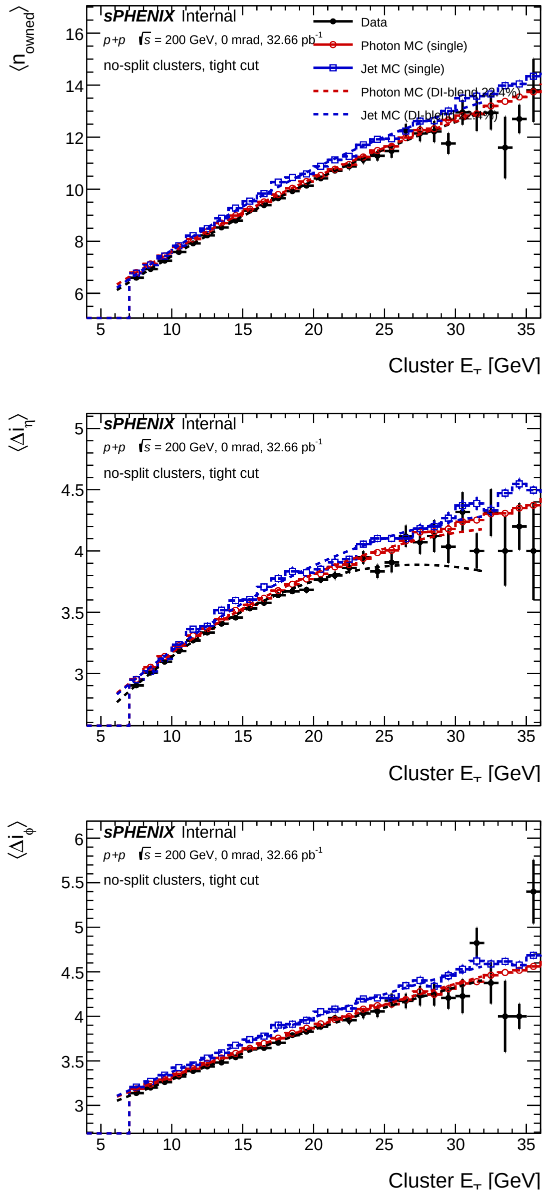


(b) 0 mrad baseline

Figure 30: Baseline cluster-size per period, split container (`CLUSTERINFO_CEMC`), data vs. pooled photon MC vs. pooled jet MC, with truth-vertex-reweighted MC. The data-jet-MC gap is larger at 1.5 mrad (+0.5 towers) than at 0 mrad (-1.1 towers); the period difference at baseline largely absorbs into the tight selection.



(a) 1.5 mrad tight (no-split)



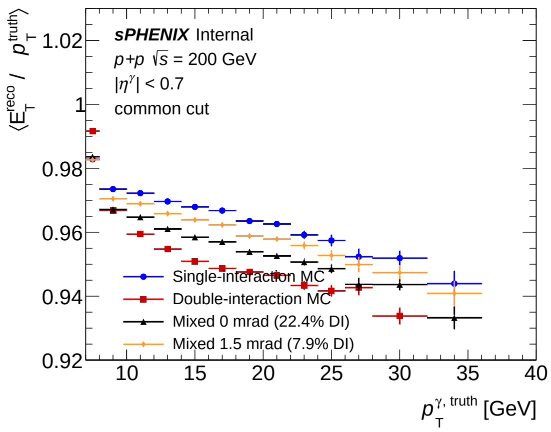
(b) 0 mrad tight (no-split)

Figure 31: Tight-selection cluster-size per period in the no-split container (`CLUSTERINFO_CEMC_NO_SPLIT`). After the tight cut, data, jet MC, and photon MC converge to the photon signal size within $\pm 5\%$ across both periods; DI effects on tight jet MC are at the sub-tower level.

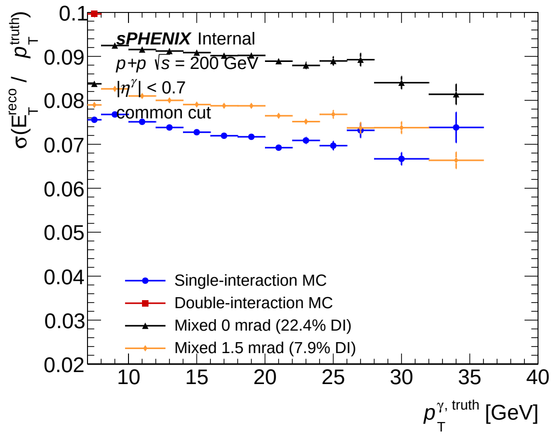
The DI effect on the ABCD background template is therefore most relevant at the *common-cut* stage, where the 22.4% jet-MC cluster-size shift feeds the shower-shape BDT distribution, and at 0 mrad more than at 1.5 mrad. After the full tight selection, the DI effect on the background template is absorbed to below the $\sim 1\%$ level on $\langle n_{\mathrm{owned}} \rangle$, which is well within the systematic budget of the background-shape cross-check.

9 Energy Response Under DI

The photon energy response $R = E_{\mathrm{T}}^{\mathrm{reco}} / p_{\mathrm{T}}^{\mathrm{truth}}$ is measured bin-by-bin in truth p_{T} across the single, double, and mixed MC, both with and without the truth-vertex reweight. The extraction is driven by `efficiencytool/compare_energy_response.C`, which builds six sample sets per p_T bin (single, double, mixed_0mrad, mixed_0mrad_vtxrw, mixed_1p5mrad, mixed_1p5mrad_vtxrw) and fits each to a double-sided Crystal-Ball (DSCB) function to extract the peak μ and resolution σ. The blended vertex scan used for the MC vertex reweight is built internally from the `photon10_nom` and `photon10_double` vertex distributions.

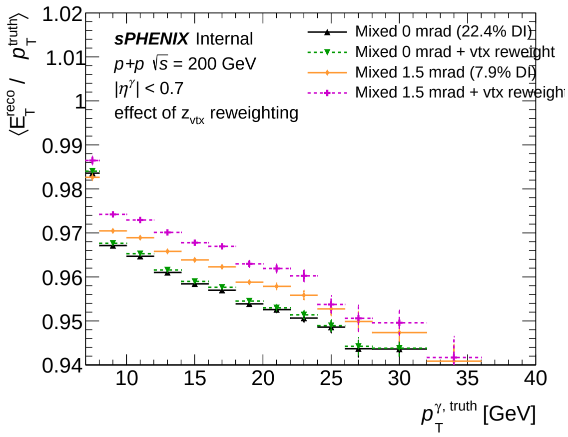


(a) Response mean μ, DI comparison

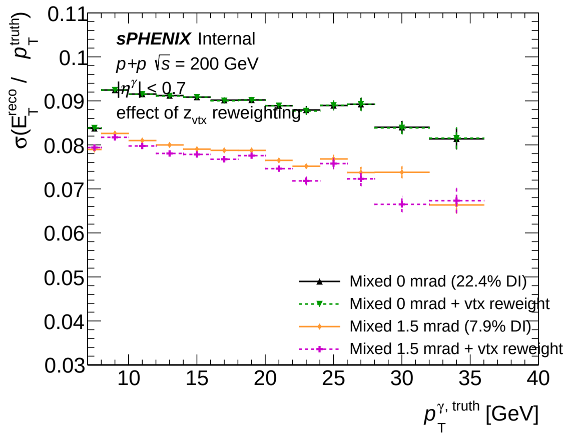


(b) Response width σ, DI comparison

Figure 32: Photon energy response $R = E_T^{\mathrm{reco}}/p_T^{\mathrm{truth}}$ at the common-cut level, across single, double, and mixed (0 mrad / 1.5 mrad) MC without the truth-vertex reweight. Left: DSCB peak $\mu(p_T)$; right: DSCB width $\sigma(p_T)$. Double MC shifts the peak by $\lesssim 0.5\%$ and broadens the width by $\lesssim 10\%$ at $p_T \sim 20\,\mathrm{GeV}$; the mixed curves sit between single and double at the appropriate fractions.

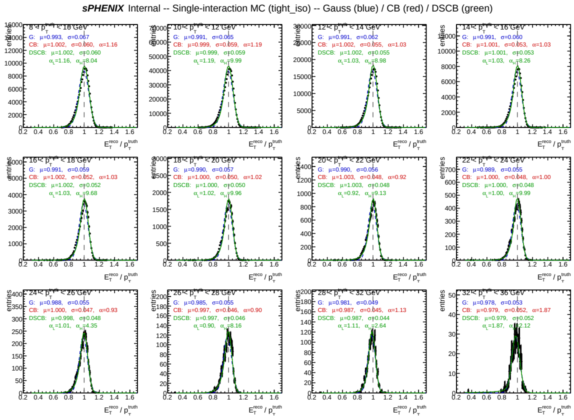


(a) Response mean μ, with truth-vertex reweight

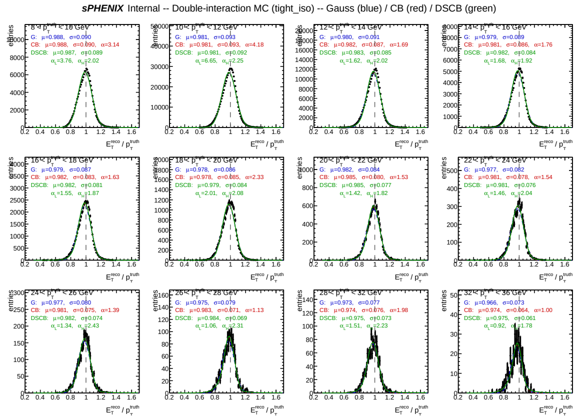


(b) Response width σ, with truth-vertex reweight

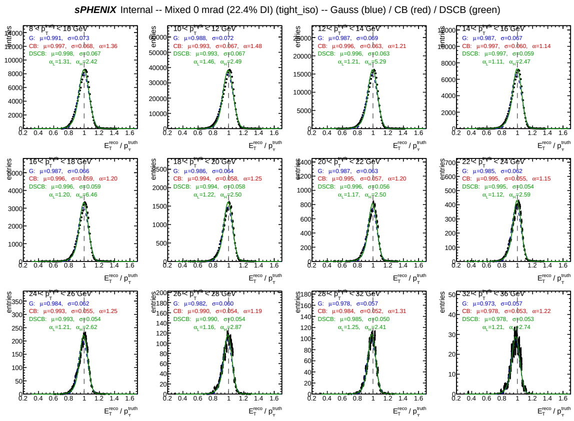
Figure 33: Photon energy response with the iterative truth-vertex reweight applied on the mixed samples (mixed_0mrad_vtxrw, mixed_1p5mrad_vtxrw). The truth-vertex reweight further reduces the residual DI-induced mean shift to below 0.3% across the full p_T range. Layout as in Fig. 32.



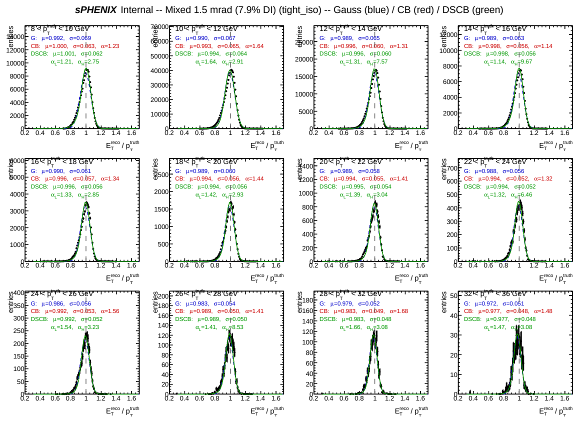
(a) Single MC, tight+iso



(b) Double MC, tight+iso



(c) Mixed 0 mrad, tight+iso



(d) Mixed 1.5 mrad, tight+iso

Figure 34: Per-bin DSCB fits to the photon energy response at the tight+iso selection level for single, double, and mixed MC. The DSCB parameters are stable across bins; the single-vs-double spread is dominated by the μ and σ shifts of Figs. 32–33.

The DI-induced peak shift of $\lesssim 0.5\%$ and width broadening of $\lesssim 10\%$ at the common-cut level sits well within the existing energy-scale systematic budget of PPG12 and is further reduced after the iterative truth-vertex reweight. The energy response is therefore not a dominant DI systematic for the cross-section measurement; the leading DI sensitivity remains on the efficiency stage (§5) and the ABCD background template (§8).

10 Eta-Edge Migration Under Pileup

The eta-edge migration study ($|\eta| < 0.7$ fiducial) exploits the fact that DI shifts the cluster η via Eq. (5), so a cluster whose truth η is just inside the fiducial edge can migrate outside the fiducial on reconstruction (producing a signal loss) and, symmetrically, a cluster whose truth η is just outside can migrate inside (producing a background fake). In the nominal 1.5 mrad period the effect is sub-leading: the vertex-shift RMS is small ($\sigma_{\Delta z} \simeq 7.6$ cm single, $\sigma_{\Delta z} \simeq 34.5$ cm double; only 7.9% of clusters draw from the wider double distribution), which translates to $\sigma_{\Delta\eta} \simeq 0.08$ in the worst case. The fiducial edge at $|\eta| = 0.7$ is $\gtrsim 5\sigma_{\Delta\eta}$ from the barrel edge at $|\eta| \simeq 1$, so the out-migration rate is percent-level. At 0 mrad the effect is proportionally larger ($\sigma_{\Delta\eta} \simeq 0.37$ averaged), but the 0 mrad period is not part of the nominal published cross-section.

The full eta-edge migration analysis (fake rates, signal leakage, ABCD R-factor bias from edge effects, net sideband flow) is documented in `wiki/concepts/eta-edge-migration.md` and the companion report in `reports/figures/` (figures `eta_migration_abcd_fakes_double.pdf`, eta_migration_matrix_double.pdf and peers). For the purposes of this DI note, the relevant observation is that the fiducial-migration effect is absorbed by the response-matrix unfolding used by PPG12 and does not need a separate DI correction. It is flagged here only because the underlying mechanism (vertex-shift-induced $\Delta\eta$) is shared with the other DI effects analysed in this note.

11 Summary, Systematics, and Recommendations

11.1 Nominal Recipe

The PPG12 DI programme converges on the following recipe for the nominal cross-section extraction:

1. **Efficiency: single-MC only, with track-ID-only matching (no $\Delta R < 0.1$ cut).** The apparent double-MC efficiency drop is dominated by a matching artifact (§5.2); removing $\Delta R < 0.1$ eliminates the artifact and reveals the genuine pileup degradation. Under track-ID matching the total photon-ID efficiency drops from 57.4% (single MC) to 52.0% (mixed 0 mrad), a $1 - 52.0/57.4 = 9.4\%$ relative reduction concentrated in the reco and BDT stages. At the PPG12 1.5 mrad working point ($f_{\mathrm{DI}} = 7.9\%$), the integrated DI effect on the cross section is 1.6% (reco) +1.0% (BDT) +0.2% (iso) = 2.8% pre-correction (see §5.6).
2. **Vertex reweighting: iterative truth-vertex $w(z)$ applied per truth vertex.** The production reco-level $f(z_r)$ structurally cannot close the individual truth vertices of DI events; the iterative method (§7) closes the truth fiducial to 1–2% inside $|z_h| < 60$ cm at 1.5 mrad and converges cleanly at 0 mrad. It removes the +1.4% cross-section bias at 1.5 mrad that the reco-level method alone could not address. The reco-level reweight remains available as a mutually-exclusive fallback; the truth-vertex variant is the nominal going forward.
3. **Shower-shape templates: mixed MC for cross-check only, not as the nominal template.** The data-vs-mixed-MC validation (§6) confirms that the mixed MC improves data-MC χ^2/ndf by up to 20.8× for pileup-sensitive observables at high E_{T}; the improvement does not hold at low E_{T} (pt0, 10–14 GeV) and the mixed MC degrades there. The analysis uses single MC for the nominal BDT evaluation and quotes the nominal-vs-mixed spread as a shower-shape systematic.
4. **ABCD background template: photon-MC-free, built from the anti-tight region in data.** The DI-induced jet-MC cluster-size shift at 0 mrad is sub-leading after the tight selection (§8); at 1.5 mrad it is negligible.

11.2 Residual Systematic

With the above recipe in place, the residual DI-related systematic at 1.5 mrad has the following ingredients:

- **Truth-vertex reweight residual**: 1–2% per bin inside $|z_h| < 60$ cm, sub-percent integrated over the analysis fiducial.
- **Efficiency pileup effect**: 2.8% integrated at 1.5 mrad (reco 1.6% + BDT 1.0% + iso 0.2%; see §5.6); an explicit single-vs-mixed-MC efficiency systematic at the cluster-weighted pileup fraction covers this contribution.
- **Shower-shape mismodelling at low $E_{\mathbf{T}}$**: picked up by the nominal-vs-mixed-MC spread on the BDT templates, explicitly quantified in the shower-shape systematic.

- **ABCD background-template DI shift**: sub-percent at 1.5 mrad after tight selection; absorbed by the background-template systematic.

- **Missing `vertexz_truth_mb` branch on current DI `bdt_split.root`**: sub-percent drop of the secondary-vertex factor $w(z_{\mathrm{mb}})$; expected sub-percent impact at 1.5 mrad and sub-percent at 0 mrad. Removed by regenerating the DI `bdt_split.root` files.

The dominant pre-existing cross-section bias at 1.5 mrad, +1.4% from the truth-fiducial mistreatment of DI events under $f(z_r)$, is eliminated by the truth-vertex reweight.

11.3 Cross-Section Impact Summary

Table 11: DI-related effects on the 1.5 mrad cross section and their treatment. Percentages are integrated over the analysis fiducial unless otherwise noted.

Effect	Pre-fix bias (1.5 mrad)	After nominal recipe	Treatment
Truth fiducial mistreatment under $f(z_r)$ alone	+1.4%	$\lesssim 0.1\%$	Iterative truth-vertex reweight
Efficiency matching artifact ($\Delta R < 0.1$)	−3.5% baseline in single MC, dominant in double MC	absorbed into nominal single-MC efficiency	Track-ID-only matching
Efficiency pileup (reco + BDT + iso combined)	2.8% (1.6+1.0+0.2)	covered by single-vs-mixed-MC systematic	Single MC nominal; mixed MC systematic
ABCD jet-template DI shift	few % on background shape at common-cut stage	sub-% on tight-selection yield	Absorbed by background-template systematic
Shower-shape low-E_{T} mismodelling	several σ on χ^2/ndf	covered by mixed-vs-nominal systematic	Nominal-vs-mixed BDT systematic
Missing $w(z_{\mathrm{mb}})$ on DI trees	—	sub-%	Regenerate DI `bdt_split.root`

11.4 Open Items

- **jet12 extension of the truth-vertex fit.** Current iterative reweight uses photon10-only. Extending to the full 4-sample blend (photon10/photon10_double + jet12/jet12_double) is straightforward but the closure numbers will move and should be remeasured.

- **DI `bdt_split.root` regeneration.** The current `photon10_double` and `jet12_double` `bdt_split.root` files predate the `vertexz_truth_mb` branch; regenerating from the reprocessed `combined.root` removes the silent secondary-vertex drop.

- **NPB purity 0 vs. 1.5 mrad split.** Deferred; the NPB-cut systematic is currently evaluated integrated over both periods. A per-period split would sharpen the pileup sensitivity.

- **Iterative-reweight speed optimization.** $\sim 10\times$ speedup available via pre-computed z^2 and dropping the per-trial mask rebuild; nice-to-have for multi-sample extensions.

- **Pipeline consolidation.** 14-way duplication of the run28 condor submit scripts + the hardcoded `total_line=10000` cap are both tracked in `condor_pipeline_review_run28.md`; neither blocks the physics at current precision.

A DI Reco–Truth Kernel

The geometric lower bound on truth-only reweights (referenced in Appendix B) can be read off the widths directly for the jet12_double sample at 1.5 mrad:

- 1.5 mrad data reco width: $\sigma \simeq 22.3$ cm
- jet12 single-interaction MC truth width (z_h): $\sigma \simeq 57.6$ cm
- jet12_double MC truth width (z_h, primary): $\sigma \simeq 62.7$ cm
- jet12_double MC reco width (z_r): $\sigma \simeq 48.6 \approx 62.7/\sqrt{2}$ (averaging)
- Conditional kernel $P(z_r|z_h = 0)$ RMS (with $z_{\rm mb}$ marginalized): $\sim$34 cm

Figure 35 shows the kernel slice directly, compared to the 1.5 mrad data target. Even if the primary truth vertex is forced to $z_h = 0$, the reco distribution from a double-interaction event cannot be narrower than the ~ 34 cm kernel width — larger than the 22 cm target. This is the structural reason the truth-only methods of Appendix B fail on DI MC, and the structural reason the iterative method of §7 needs the $w(z_h)\,w(z_{\rm mb})$ factorization rather than a w-on-z_r construction.

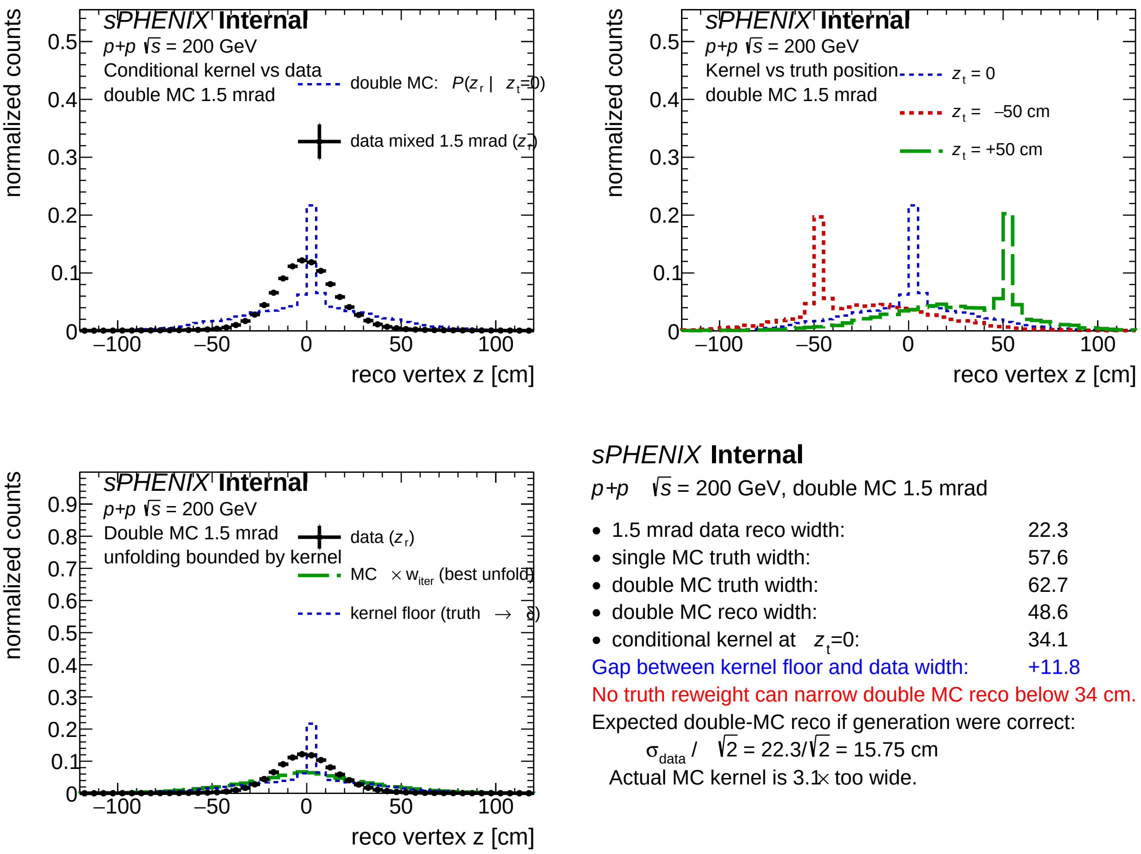


Figure 35: Conditional reco kernel $P(z_r|z_h)$ in jet12_double MC at 1.5 mrad (with $z_{\rm mb}$ marginalized), with the 1.5 mrad data reco z_r shape overlaid. The kernel's ~ 34 cm RMS exceeds the $\sim$22 cm data peak, which is the geometric lower bound on truth-only reweighting.

B Discarded Reweighting Methods

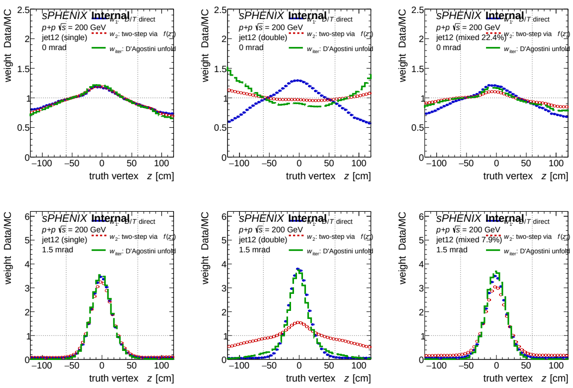
Before the iterative method of §7 was settled on, three truth-only reweight constructions and one closed-form 2D construction were explored. All four were discarded; the essentials:

w_1**: direct ratio.** $w_1(z_h) = D(z_r)/T(z_h)$ with the common binning identifying z_r bins with z_h bins. Breaks down on the DI sample because $P(z_r|z_h)$ is wide: the LHS has a narrow data target while the RHS integrates over a wide kernel.

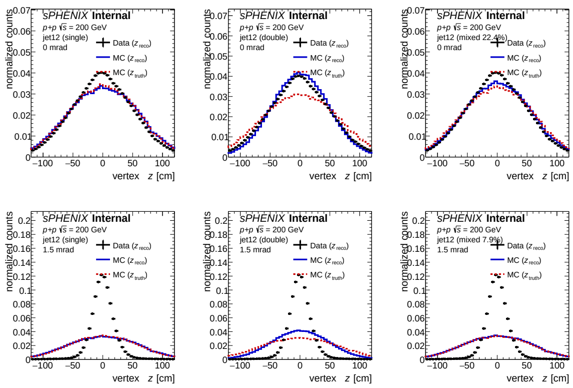
w_2**: conditional projection.** $w_2(z_h) = \mathbb{E}[f(z_r) \,|\, z_h]$, the expectation of the production reco weight conditional on the primary truth vertex. Also bounded from below by the DI kernel width (Appendix A).

$w_{\text{iter,DA}}$**: D'Agostini iterative unfolding.** Iterative Bayesian unfolding using the MC response matrix $R_{ij} = P(z_r^i|z_h^j)$. Hits the same geometric floor at 1.5 mrad — the DI reco-truth kernel is too wide to invert within the data-domain support.

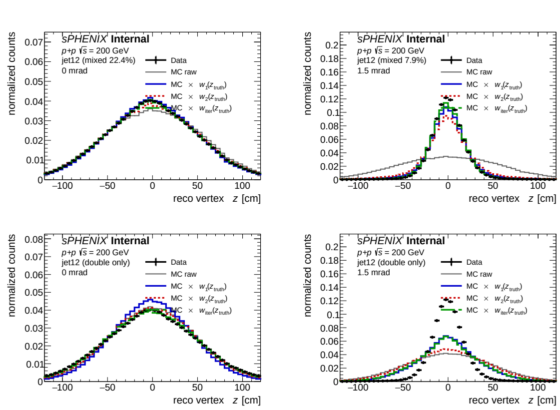
Closed-form 2D Gaussian reweight. $w(z_h, z_{\text{mb}}) \propto \exp[-\frac{1}{2}(z_h^2 + z_{\text{mb}}^2)\,\Delta]$ with $\Delta = 1/\sigma_{\text{lum}}^2 - 1/\sigma_{\text{MC}}^2$. This acts on both the symmetric and antisymmetric coordinates simultaneously and closes all three vertex axes; however, it assumes a single Gaussian shape for the data luminous region. At 1.5 mrad the data has a ±-asymmetric mid-tail bump that a symmetric Gaussian cannot reproduce, so the closed form was superseded by the iterative method that discovers $w(z)$ from the data directly without any Gaussian assumption.



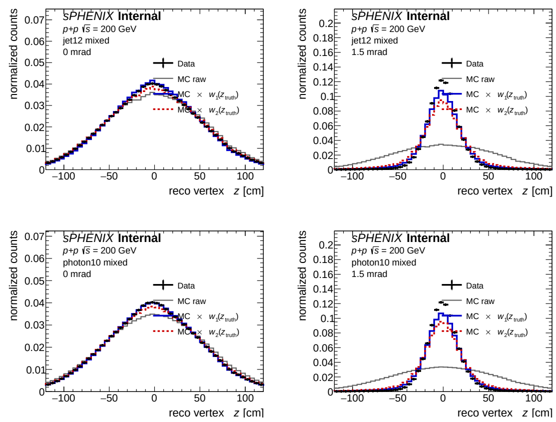
(a) Recovered $w(z)$: discarded methods



(b) Truth-vertex shapes: discarded methods



(c) Closure: discarded methods



(d) Jet vs. photon closure: discarded methods

Figure 36: Diagnostic plots for the discarded truth-only reweight methods. All four methods hit a geometric floor on the DI sample at 1.5 mrad set by the reco-truth kernel width.

C TruthVertexRecoCheck Baseline

Before building any truth-based reweight, a sanity check on the effect of dropping MC events whose reco MBD vertex is the -9999 sentinel was performed. The concern: an MC event that loses its reco vertex may be physically unusual (e.g. with a shifted truth vertex z_h), so dropping it could skew the efficiency in a z_h-dependent way.

Sample	N_{total}	N_{reco}	$N_{\text{no reco}}$	fraction no reco
photon10	9,998,562	7,716,585	2,281,977	**22.82%**
photon10_double	9,997,075	9,491,223	505,852	**5.06%**
jet12	10,000,000	7,689,530	2,310,470	**23.10%**
jet12_double	9,997,000	9,485,384	511,616	**5.12%**

Table 12: Fraction of events without an MBD reco vertex. The double-interaction productions have $\sim 4.5\times$ higher MBD vertex-finding efficiency than their single counterparts (two concurrent collisions give the MBD twice the chance of producing a vertex).

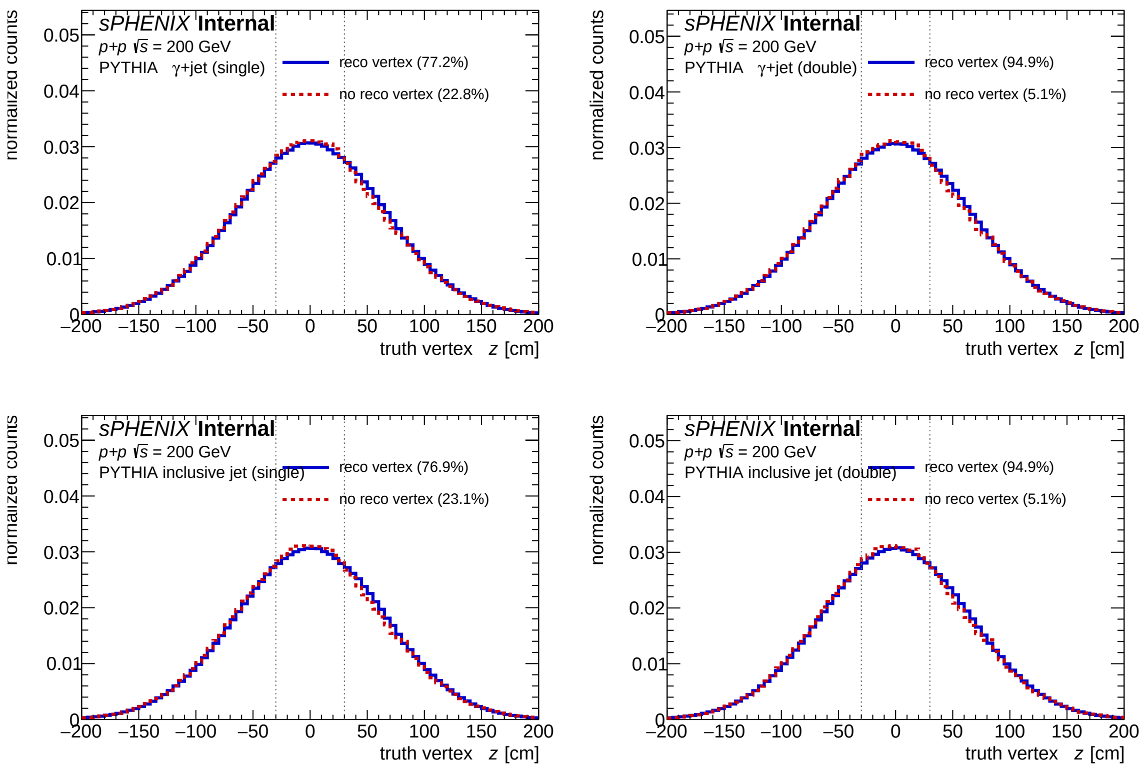


Figure 37: Primary truth vertex z_h distribution for events with (blue) and without (red dashed) a reconstructed MBD vertex z_r, area-normalized for shape comparison. Columns: single / double interaction. Rows: photon10 / jet12. Dotted vertical lines: $|z_h| = 30\,\text{cm}$ reference.

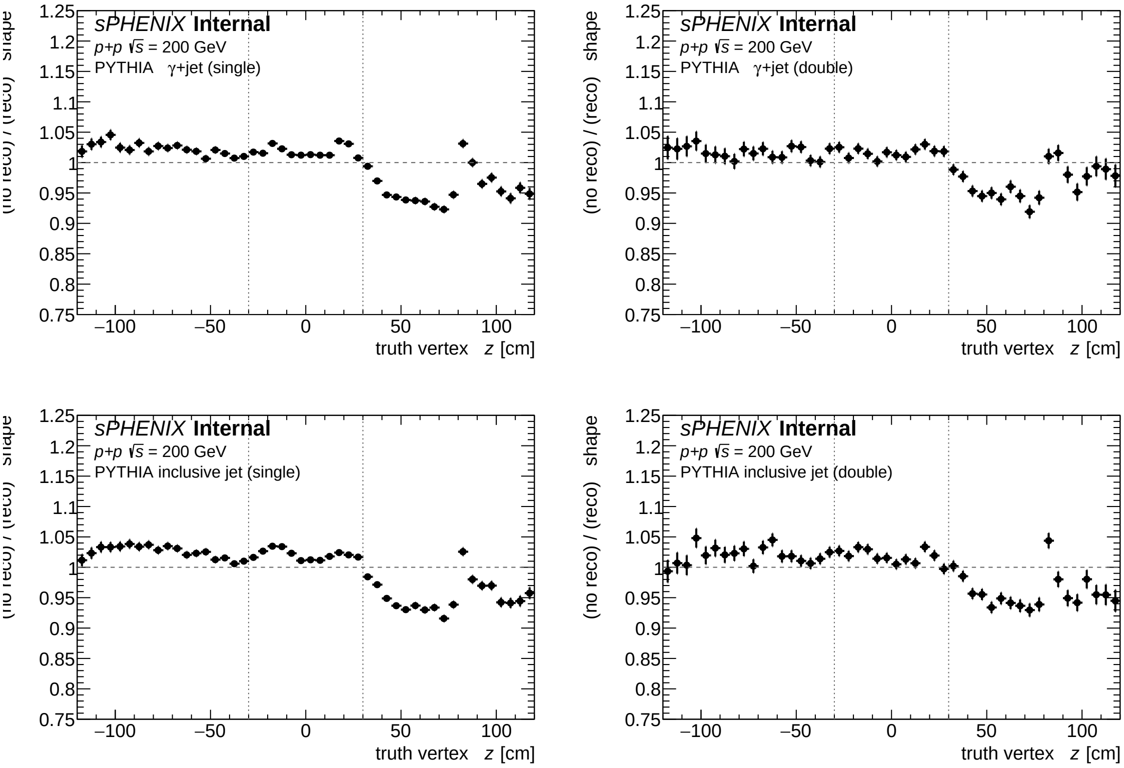


Figure 38: Ratio (no-reco-vertex shape)/(reco-vertex shape) vs. z_h, zoomed to $[0.75, 1.25]$. All samples show a $\sim \pm 5\%$ north/south asymmetry inside $|z_h| < 30\,\mathrm{cm}$, consistent with a small MBD geometric bias.

Inside $|z_h| < 30\,\mathrm{cm}$ the two shapes agree to within $\pm 5\%$ with a small z_h-asymmetry that is the same in all four samples and therefore a detector-level effect (MBD vertex finder slightly more efficient at positive z). Dropping the no-reco-vertex events is *approximately* harmless — both the efficiency numerator and denominator are scaled by the same factor, and the residual z dependence of the drop rate is small. However, dropping $\sim$22% of the single-interaction MC sample is still a lot of statistics, and it makes the efficiency measurement implicitly conditional on a valid reco vertex being found. That was the original motivation for exploring a truth-level reweight in the first place.

D File and Code Inventory

Code.

- `anatreemaker/source/CaloAna24.{h,cc}` — writes the `vertexz_truth` (z_h) and `vertexz_truth_second` (z_{mb}, read in C++ as `vertexz_truth_mb`) branches on the slimtree.
- `efficiencytool/ShowerShapeCheck.C` — shower-shape analysis with the `mix_weight` + `vtxscan_sim_override` DI-blend options.
- `efficiencytool/run_showershape_double.sh` — two-pass full-GEANT blending pipeline.
- `efficiencytool/RecoEffCalculator_TTreeReader.C` — efficiency calculator (`trkID`-only matching; `SetUseWeightedEvents` fixed).
- `efficiencytool/compare_energy_response.C` — energy response $\mu(p_{\mathrm{T}})$, $\sigma(p_{\mathrm{T}})$ under DSCB across single/double/mixed MC with optional vertex reweight.

- efficiencytool/truth_vertex_reweight/ — standalone Python tool for the iterative method:
  - fit_core.py — cached sample loader, mixed_mc_histogram forward model, Poisson χ^2.
  - fit_truth_vertex_reweight.py — driver producing reweight.root and fit_result.yaml.
  - plot_closure.py, plot_comparisons.py — closure and cross-period diagnostic plots.
  - config.yaml — sample paths, run ranges, fit parameters.
- efficiencytool/TruthVertexRecoCheck.C — Appendix C supporting macro.
- efficiencytool/calc_pileup_range.C — MBD-based pileup-fraction calculator.
- efficiencytool/MbdPileupHelper.h — MBD pileup metric helper (avg / prod / max σ_t etc.).
- efficiencytool/DoubleInteractionCheck.C — toy vertex-shift simulation (not part of the nominal DI treatment; see .claude/rules/double-interaction.md).

Config fields.

- truth_vertex_reweight_on (0/1): enables the iterative truth-vertex reweight.
- truth_vertex_reweight_file: path to reweight.root containing h_w_iterative.
- vertex_reweight_on (0/1): legacy reco-level $f(z_r)$; mutually exclusive with truth_vertex_reweight_on.
- vertex_scan_data_file: unrestricted data vertex distribution for the toy sim (DI cross-check only).

Output files.

- efficiencytool/truth_vertex_reweight/output/{1p5mrad,0mrad}/reweight.root — per-period $w(z)$ and diagnostic histograms.
- efficiencytool/truth_vertex_reweight/output/{1p5mrad,0mrad}/fit_result.yaml — fit metadata and convergence statistics.
- results/MC_efficiencyshower_shape_{sample}_{suffix}_vtxscan.root — Pass-1 blended vertex distribution.
- results/MC_efficiencyshower_shape_{sample}_combined[_inclusive]_{suffix}.root — Pass-2 merged full-analysis output.

Known source-plot cosmetic issues (reused as-is). Two classes of known source-PDF cosmetic issues are reused from the upstream plotting code without post-editing (this note preserves the existing figures verbatim rather than regenerating them):

- The per-bin DSCB response grids in §9 (Fig. 34) can show slight legend/data overlap in the top-left panel on some pages; this is a matplotlib auto-layout artifact in the upstream compare_energy_response.C grid plotter and does not affect the numerical fits.
- The truth-vertex iterative-closure and method-comparison figures (Figs. 20–25) carry Python fig.suptitle() banners inherited from truth_vertex_reweight/plot_closure.py. These banners duplicate information already captured in the LaTeX captions but are kept unedited to preserve byte-level reproducibility against the truth_vertex_reco_check.pdf source report.

These are flagged here for transparency and do not reflect errors in the physics content.